

Credit Airlines: Financing Transit Lags, Export Capacity and Comparative Advantage*

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Abstract

How does access to finance interact with transportation modes to shape export performance? We link Colombian customs export data, central bank payment data, and firm-bank loan data. We show that sea freight is cheaper than air but yields payments 2 to 3.5 months slower, thus requiring more working capital. Causal estimates confirm that improved bank credit access leads exporters to substitute toward cheaper, slower sea shipping. Embedding mode choices and working-capital costs into a quantitative multi-sector trade model, we show that financial development expands aggregate export capacity and benefits sea-intensive sectors. Mode-specific payment lags account for over 50% of these export capacity gains and 10% of shifts in sectoral comparative advantage.

Keywords: transportation, access to finance, comparative advantage

JEL Codes: F12, R40, F65

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1 Introduction

There is a growing literature showing that transportation decisions are central to international market access and trade patterns. Much less is known about how transport-mode-specific shipping lags affect firms’ working-capital needs, how cost of finance shapes the choice of transport mode, and how this margin feeds into exporting capacity and sectoral specialization. Recent episodes in global shipping, particularly in emerging economies, have highlighted that slower maritime shipments can lock up working capital for months, especially for those with less access to external financing.¹

We use a novel dataset that links customs export shipment records, firms’ financial statements, banking relationships, and export payments data for Colombia. We show that transactions shipped by air incur higher freight costs but face systematically shorter payment lags than those shipped by sea.² This implies higher working-capital needs for maritime shipments. Building on this finding, we show that the probability of shipping by air decreases with better access to banking, highlighting the role of finance in transport-mode choice. Firms can partially substitute credit with air shipment, but at the cost of higher air freight costs - hence our title “credit airlines”. As financial access improves, the working capital needs caused by longer sea delivery lags become less costly, decreasing effective trade costs and increasing total exports. Moreover, since sea intensity varies by sectors, access to finance shifts comparative advantage towards sea-intensive sectors.

The central mechanism of our paper is the differential in timing of payments across transport modes. Firms that ship by sea wait longer to be paid simply because their goods spend more time in transit, so each shipment ties up working capital for longer than an equivalent air shipment. This longer financing horizon can make fast air freight relatively more attractive as a way to shorten the cash-flow cycle. In this environment, better access to bank credit relaxes working-capital constraints and allows firms to tolerate slower payment cycles and export more. Further, shipping choices shift toward sea freight and ultimately affect which sectors can profitably serve international markets.

We develop a trade model where firms need external financing to cover working capital costs for both production and shipment time, where the shipment time depends on the mode

¹For instance, the COVID-19 Pandemic (2020–2021) generated severe port congestion and container shortages that added weeks to many sea shipments. Another example is the Suez Canal Blockage (March 2021), which forced ships to wait or reroute around the Cape of Good Hope, delaying deliveries by up to several weeks. Finally, the Red Sea Crisis (2023–ongoing) has led carriers to reroute traffic away from the Red Sea, lengthening Asia–Europe sea routes by roughly 10–14 days. In all three cases, longer and more uncertain sea transit and port times extended the period between shipment and payment, tying up firms’ working capital in goods stuck at sea or in congested ports.

²Higher payment lags do not imply delinquency, our statement is only about the lag between when the shipment leaves Colombia and when the payment is received.

of transport. We calibrate the model to Colombia and use it to quantify how cheaper external finance affects modal choice and comparative advantage. Lowering the average Colombian interest rate increases exports, decreases air share and shifts the comparative advantage of the country towards sea-intensive sectors and towards sectors with high financing need in production.

Empirical facts In the first part of the paper, we leverage our detailed Colombian administrative data from 2008 to 2019 to document three facts.

First, we show that air shipping costs are more expensive than sea shipping costs, but exporters receive faster payments when shipping by air. Using the freight cost share of export value from the customs data, we find an air premium of around 5.5 percentage points after controlling for a stringent good-origin-destination-time fixed effect. Using our data on export payments, we show that firms shipping by sea face a larger gap between shipment and payment. We use firm-month data on exports and associated payment inflows to measure payment delays relative to the last shipment. We find that firms shipping by air receive payments on average about 2 to 3.5 months earlier than firms shipping by sea, even controlling for time, good and destination. The difference disappears once controlling for the shipping time to the destination, suggesting that longer delivery lags are driving the payment lag. Although this is a common assumption ([Amiti and Weinstein, 2011](#)), ours is the first paper to document this fact in the data to the best of our knowledge.

Second, we show that better access to bank credit leads firms to substitute away from fast, expensive air freight toward slower, cheaper sea freight. Using shipment-level data linked to firm-level credit, we estimate how banking access — measured as interest payments to sales — affects the probability that an export is shipped by air, controlling for shipment value, rich origin-destination-good-time fixed effects, and lagged firm characteristics. Our results are robust to alternative measures of banking access. To disentangle credit supply from credit demand, we instrument credit access with firms’ exposure to banks’ Tier 1 capital ratios, a standard measure of core loss-absorbing capacity that proxies banks’ ability to sustain lending ([Correia et al., 2024](#); [De Jonghe et al., 2020](#); [Jiménez et al., 2017](#); [Gropp et al., 2019](#); [Ormazabal et al., 2025](#)). We find that a one-standard deviation increase in interest-payments-to-sales in this IV framework reduces the probability of air shipping from 51% to 23%, which, in back-of-the-envelope terms around the sample average, implies transport cost savings of about 1.7% of annual exports.

Third, we document that air intensity varies by good, although many products are shipped by both air and sea. About 15% of goods never use air, around 10% always use air. Air share ranges from near 0 for bulky, low-value items to 100% for high-value or highly

perishable goods. Thus, access to finance affects not only the choice of shipping mode as documented in our second fact, but also sectoral specialization. Better access to finance decreases working capital costs associated with longer sea shipping lags. This disproportionately benefits sectors that are sea-intensive, thereby affecting sectoral comparative advantage at the aggregate level as well.

Quantitative framework and counterfactual To quantify the impact of financial access on export volume, aggregate air usage and on comparative advantage, we develop a two-country international Krugman model with multiple sectors, working capital constraints and choice of transportation modes. The framework of our model captures our three empirical facts. Sector-mode specific trade costs and taste shifters drive heterogeneity in sectoral air intensity. Firms need to pay their factor of production immediately but shipments are paid only upon delivery, so they need to borrow and incur interest costs until delivery. Delivery times are mode-specific and firms choose their shipping mode optimally. In doing so, they face a trade off between higher interest cost if they ship by sea versus potentially lower iceberg shipping cost. A tractable discrete choice model delivers a key mode-switching elasticity that governs how a change in relative mode shipping costs –including time financing cost– translates into modal share. The model can be aggregated at the sectoral level such that the aggregate sectoral export cost depends on all modes’ specific iceberg costs, delivery lags and interest rate.

Sectoral credit access is modeled through an exogenous interest rate.³ A decrease in interest rate decreases the share of air within a sector and decreases overall export costs. In particular, export costs of sectors that are relatively sea-intensive – for example bulky items with a lower relative cost of sea – decrease more. As a consequence, sectoral composition of exports tends to shift towards sea-intensive sectors.

Sectors also differ in their overall time-to-build and hence have heterogeneous exposure to financing cost even for domestic production and regardless of shipping decisions. Hence, change in sectoral composition is driven by two channels. The first channel is well known: reduction in overall financial cost shifts comparative advantage towards finance-dependent sectors (Manova, 2008).⁴ The second channel is novel: reduction in the financial cost of longer sea shipment lags increases the volume of exports and shifts comparative advantage

³We model access to finance following Ramos-Menchelli and Van Doornik (2023) who consider that credit monitoring costs are a deadweight loss. We abstract from modeling access to finance channels such as trade finance, loan guarantees, letter of credit, and insurance. For more details on how these issues impact exporters see Garcia-Marin et al. (2023), Niepmann and Schmidt-Eisenlohr (2017a), Niepmann and Schmidt-Eisenlohr (2017b), and Schmidt-Eisenlohr (2013)

⁴We abstract away from firm heterogeneity and the well-studied distinction between extensive and intensive margin (Manova, 2013).

towards sea-intensive sectors.

To discipline the model, we use the customs data to compute trade shares and mode shares. We estimate the elasticity of substitution across mode in a theory-consistent regression. We regress sectoral mode share on sectoral interest rates, using variation in interest rate across sectors and time. The resulting coefficient has the structural interpretation of the mode elasticity combined with the difference in payment lag between air and sea. Finally, we use firm balance sheet data to match sectoral debt-to-sales ratio as a way to discipline the sectoral heterogeneity in external finance exposure in production.

Our main counterfactual lowers the average Colombian interest rate to the level of the US interest rate — an approximately 7 percentage points decrease. This interest rate change results in an increase in aggregate exports by around 4.2%, with the payment lag channel accounting for more than half of the total increase. Moreover, value-weighted air share decreases by 1.2pp out of a baseline of 26%. The majority of modal change happens within sector, with sea shipment becoming relatively cheaper than air due to reduced financing cost. A smaller part is due to reallocation towards sea-intensive sectors and comparative advantage shifting towards sea-intensive sectors. The calibration implies that our novel channel accounts for about 10% of the variation in sectoral revealed comparative advantage in the counterfactual, while the rest is driven by sectoral variation in external financing exposure.

We also use our model to compute the change in CO2 emission following improved financial access. Two effects go in opposite directions: first, emissions increase because of higher export volumes. Second, they decrease because of the shift towards the less polluting maritime mode. Our calibration implies that CO2 emissions can decrease even though aggregate exports actually increase.

Related literature This paper contributes to three strands of literature. First, it adds to work on the link between access to finance and international trade by using novel data on export payments that reveal the payment lags faced by sea exporters and by providing causal evidence that better access to bank finance reduces the probability of shipping by air. Within this literature, this paper is closely related to studies of how credit shocks and financial frictions shape the performance of exporting firms (Matray et al., 2024; Paravisini et al., 2023; Friedrich and Zator, 2023; Xu, 2022; Feenstra et al., 2014; Amiti and Weinstein, 2011; Minetti and Zhu, 2011).⁵ We also contribute to the trade credit literature, which studies how firms use supplier credit to finance transactions when financial frictions are

⁵Relative to Paravisini et al. (2015), who do not find a significant response of the share of exports shipped by air to credit-supply shocks in annual firm-level data, we use granular transaction-level information on mode of transport, which allows us to uncover sizable effects of credit supply on firms' use of air freight.

present (Hardy et al., 2022; Chod et al., 2019; Fabbri and Klapper, 2016; Schmidt-Eisenlohr, 2013; Fisman and Love, 2003; Wu et al., 2014); here, we highlight the additional financing needs borne by the firm that extends trade credit to cover longer payment lags depending on the mode of transportation.

Second, we contribute to the literature on transportation and international trade by introducing external finance as a determinant of modal choice. A large body of work studies how transportation costs shape trade patterns (Ganapati et al., 2024; Ducruet et al., 2024; Baldomero-Quintana, 2025; Bonadio, 2024), and air shipment specifically has been shown to serve as a way to trade time for money (Hummels and Schaur, 2013; Harrigan, 2010). Recent work has documented that firms substitute between air and sea depending on product characteristics and trade costs (Tolva, 2024; Lugovskyy et al., 2025), or use multiple modes jointly (Fuchs and Wong, 2024). We show that in addition to goods-specific physical characteristics and monetary costs, this substitution reflects another economic tradeoff: air shipping is more expensive in monetary terms, while sea shipping is more expensive in terms of credit. As a result, firms can substitute credit with air shipments, meaning that financial conditions are an additional margin governing modal substitution. Since we provide evidence that modal choice and access to finance jointly drive trade patterns, our manuscript represents a bridge between the canonical work of Harrigan (2010) and Manova (2013).

Last, our paper contributes to the macro-development literature on financial frictions and comparative advantage (Leibovici, 2021; Besedes et al., 2014; Beck, 2002; Manova, 2013, 2008; Kletzer and Bardhan, 1987). Building on work that links financial development to countries' specialization patterns, we show that access to finance also operates through firms' transport-mode choices and the associated payment delays, shaping comparative advantage through two channels. The first is well established: lower financial costs shift comparative advantage toward sectors with greater external finance dependence (Manova, 2008). The second is novel: because sea shipping entails longer payment lags than air and therefore higher working-capital requirements, lower financial costs also shift comparative advantage toward sea-intensive sectors.⁶ Our model captures both channels and allows us to isolate each one transparently.

We proceed as follows. Section 2 describes the data. Section 3 presents the empirical evidence. Section 4 develops the theoretical model. Section 5 reports the main counterfactual results. Section 6 concludes.

⁶Our analysis focuses on how financial development shapes comparative advantage through transport-mode choice and payment delays, abstracting from potential feedback effects running from specialization patterns to financial development (Do and Levchenko, 2007).

2 Data and Summary Statistics

We use four administrative data sources on Colombian firms from 2008 to 2019, focusing on exporting firms that we link to their export payments, financial statements, and bank credit. Colombia is a mid-sized open economy whose export basket spans commodity-intensive sectors — mineral fuels, coffee, and cut flowers — and whose firms rely primarily on bank credit for external financing, with corporate borrowing rates substantially higher than in advanced economies and broadly in line with the Latin American average. Appendix A.1 provides a detailed description of the export structure and financial system.

A key feature of our setting is novel transaction-level data on export payments, which allow us to observe the timing and structure of foreign-currency inflows associated with exports.

- **Customs.** Our dataset contains all international trade transactions between 2008 and 2019 for Colombia. The source is the Colombian tax and customs office, (*Dirección de Impuestos y Aduanas Nacionales* or DIAN). For each transaction, we observe the good, weight, units, date, mode of transportation, firms, the city of origin, the city of destination, the value of the shipment, and the freight cost of shipment.
- **Export payments.** We use exports' payments data (*Formato 2, Banco de la República de Colombia*). Banco de la República records these data when registering foreign-currency inflows into Colombia associated with export transactions. For each observation, we see every payment received by an exporting firm, explicitly labeled as a payment for exports and classified as an advance payment, a payment made within one year of the export date, or a payment made more than one year after the export.
- **Bank credit.** We use data from the Colombian credit registry (*Formato 341*). In this dataset, we observe exporting and non-exporting firms' loan characteristics such as interest rate, maturity, total balance, interest payments, and the loan openings' dates.
- **Financial statements.** We obtain firm's financial reports from the *Superintendencia de Sociedades*, a Colombian government agency that oversees and regulates large private Colombian corporations.⁷ The private firms in this dataset are among the largest in Colombia.⁸ We use information on balance sheet characteristics such as sales, leverage, cash-to-assets, inventory-to-assets and the sector of every firm.

⁷See appendix A for further details on exports data, and [Aristizábal-Ramírez and Posso \(2021\)](#) for further details on firm's balance sheet and credit registry

⁸The average sales of the Colombian firms in this dataset amount to 10 million USD. This reflects that the firms we consider are large, given the size of the Colombian economy.

We use these sources to construct two complementary datasets. First, we build a firm-month panel that combines exports and associated payment inflows to study the timing of export payments relative to shipment dates.

One drawback of our data is that while we observe total incoming international payments for each month at the firm-level, they are not matched to a specific transaction. Hence, we collapse our data to the monthly level and focus on isolated export shipments, where we can safely attribute a payment to a transaction. We restrict the sample to payments made from the month of the first shipment onward, and to payments following any shipment that occurs at least one year after the previous shipment. Further, we drop shipments that are followed by another shipment in the same year. We argue that those restrictions guarantee that the observed payment relates to the last observed shipment.⁹

Panel A of table 1 summarizes this dataset: it shows that firms receive short-term (within one year of the shipment) payments with a relatively higher probability, with a lower probability for long-term payments (over one year). Firms, on average, receive advance payments, but the mean payment arrives about 11 months after the last shipment, with substantial dispersion as the median payment occurs within one month, and shipment values—mostly transported by air—range from roughly \$1,000 to \$414,000.

Second, we merge export transactions with annual firm-level financial statements and bank credit information. The resulting dataset covers 4,314 exporting firms with both balance sheet and credit registry data and includes 4,771,563 export transactions, which we use to study how access to bank credit shapes the probability of shipping by air. The remaining panels of Table 1 summarize this dataset: the average shipment is valued at about 30,000 USD, 52% of transactions use air freight, average annual sales per firm are roughly 30 million USD (around three times the average for all Colombian firms), the mean leverage ratio (total liabilities over total assets) is 0.37, and firms face an average interest rate of 7.9 percent, approximately 0.6 percentage points below the economy-wide corporate loan rate over the same period.¹⁰

3 Empirical facts

In this section, we show three empirical facts regarding access to finance and exporters' choice of mode. First, we show that air shipping costs are more expensive than sea shipping

⁹Appendix A.5 describes in detail how we construct the firm-month panel and the estimation samples used in Table 3 and Appendix Table ??.

¹⁰According to the Superintendencia Financiera, average effective annual interest rates for corporate loans over our sample period (2008–2019) were approximately 15.54% for ordinary commercial loans, 10.35% for preferential commercial loans, and 9.69% for treasury (short-term working-capital) loans.

Table 1: Summary Statistics

	Mean	Std. Dev	Pctile. 5	Median	Pctile. 95	N
A. Exports and Payments, monthly						
Prob. of receiving advanced payments	0.66	0.47	0	1	1	1162894
Prob. of receiving short-term payments	0.86	0.34	0	1	1	1162894
Prob. of receiving long-term payments	0.17	0.37	0	0	1	1162894
Months since last shipment	11	14	0	4	41	1162894
Last export by air (dummy)	0.60	0.48	0	1	1	1162894
Export value of last transaction ^(k)	153	1166	1	14	414	1162894
B. Export transactions						
Unit value (USD per unit)	253	13936	0.13	4.3	131	4559233
Air (dummy)	0.52	0.50	0	1	1	4559233
Export value ^(k)	28	264	0.04	2.2	79	4559233
Modal share of export value	0.5	0.48	0	0.49	1	297634
C. Financial Statements						
Sales ^(m)	31	121	0.66	5.9	126	19662
Leverage (ratio)	0.37	0.24	0.05	0.35	0.76	19658
Inventory to assets (ratio)	0.21	0.15	0.01	0.19	0.49	19658
Cash to assets (ratio)	0.06	0.08	0	0.03	0.22	19658
Sales growth (%)	6.9	364	-35	1.4	43	19662
FDI receipt (dummy)	0.05	0.22	0	0	1	19736
D. Credit Registry						
Int. payments to sales (%)	0.79	2.7	0	0.2	2.8	19736
Total debt ^(m)	13	50	0.01	1.5	59	19736
Number of active loans	8.6	7.0	1	7	23	19736
Int. rate	7.9	6.3	1.1	6.3	22	17799
Instrument	0.76	0.82	0	0.57	2.1	19736
Tier 1 capital ratio	17.06	12.39	10.60	14.09	32.82	274

Note: This table reports summary statistics at the firm, export transaction, and monthly export-relationship levels. Superscript ^(k) denotes variables measured in 2018 USD thousands; superscript ^(m) denotes variables measured in 2018 USD millions. Pctile. 5, Median, and Pctile. 95 denote the 5th percentile, the median, and the 95th percentile of each variable, respectively. The instrument corresponds to the weighted average of a firm's exposure to banks with stronger Tier 1 capital ratios. Source: *SuperSociedades*, *DIAN*, *Formato 341*, *Banco de la República* and author's elaboration.

costs, but transactions shipped by sea face systematically longer payment lags than those shipped by air, implying higher working-capital needs for maritime shipments. Second, we show that better access to bank credit leads firms to substitute away from fast, expensive air freight toward slower, cheaper sea freight. Third, we show that different goods have different air intensity.

Fact 1: Air freight rates are more expensive but air shipping leads to faster payments

Air cost premium We first document that shipping costs are larger for air shipments than sea shipments. While it is well-known that the price per kilo and kilometer is higher on average, the raw difference reflects many confounders — shipments by air tend to involve higher-value goods, different routes, and time-sensitive cargo that might have higher freight costs regardless. Using freight charges in our export data, we compute each shipment's cost

share of export value and regress this ratio on a dummy for air shipment, controlling for stringent fixed effects:

$$\mathbf{Cost}_{i,odgt} = \beta_1 \mathbf{Air}_i + \alpha_{odgt} + \varepsilon_{i,odgt}, \quad (1)$$

where $\mathbf{Cost}_{i,odgt}$ is the freight cost to export value ratio for shipment i originating from department o to country d for a good g in year-month t . The inclusion of the fixed effect α_{odgt} implies that we compare shipments of the same good made in the same month, from the same origin to the same destination.

Table 2 reports the results of estimating regression (1) for various specifications of fixed effect, including additional firm fixed effects. Overall, the results show a stable coefficient of around 0.055 for the **Air** dummy. This implies that shipping by air is subject to an “air premium” of around 5.5 percentage point within the same origin-destination-good-time combination.

Table 2: Cost Premium of Air Shipping

Transportation Cost to export value			
	(1)	(2)	(3)
Air	0.061** (0.029)	0.053* (0.030)	0.056* (0.031)
Observations	7126015	7123175	7321494
odt_hs8 FE	✓	✓	
odt_hs6 FE			✓
Firm FE		✓	✓

Note: This table reports the cost premium of air shipping from equation 1. odt_hs8 denotes origin region, destination city, month, year, and HS8 product classification; odt_hs6 denotes the same at the HS6 product level. Standard errors clustered at the firm level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Air payment timing differential We provide evidence that firms that ship by sea experience a larger gap between shipment and payment, which can increase their reliance on external financing and imply higher working-capital needs for maritime trade.

For an export transaction, we compute the time delay between the month of the transaction and the first payment thereafter. We focus on payments occurring after the very first export of the firm to ensure we match the payment to the right export transaction.¹¹ We then compare this delay for transactions by air or sea, controlling for the size of the

¹¹Appendix Table B.3 repeats the exercise including all payments that are more than 12 month after and 12 months before the previous or successive payment, finding similar patterns.

transaction and a variety of fixed effects. To do so, we estimate the following regression:

$$\text{delay}_{it} = \alpha_t + \beta_1 \text{Air}_{it} + \beta_2 \ln \text{value}_{it} + \delta_{\text{gap}(i)} + \gamma_{g(i)} + \eta_{d(i)} + \varepsilon_{it}, \quad (2)$$

where i is an incoming international payment, delay_i is the difference, in number of months, between a payment and the date of the last shipment. Air_i is a dummy for whether the export was made by air.¹² In some specifications, we add fixed effects for the gap between the matched export transaction and the previous or next export transaction, $\delta_{\text{gap}(i)}$. We also control for a good (HS-6) fixed effect ($\gamma_{g(i)}$), destination fixed effect ($\eta_{d(i)}$) and month fixed effect (α_t). We weight each observation by value, so that the coefficient can be interpreted as the difference in average number of months taken to receive a dollar for an air shipment relative to a sea shipment. This estimated differential should be interpreted as a reduced-form cash-recovery lag associated with each transport mode, reflecting not only physical transit and logistics but also the typical contractual, institutional, and trade-finance arrangements that accompany air versus sea shipments, which we treat as stable at the time of shipment.

Table 3: Air shipments are paid around 3.5 months earlier

	Time since shipment (delay_i)			
	(1)	(2)	(3)	(4)
Air_i	-3.456*** (0.593)	-2.905*** (0.576)	-2.078** (0.889)	0.495 (1.246)
$\ln \text{value}_i$	-1.615*** (0.121)	-1.370*** (0.121)	-0.807*** (0.251)	-0.527** (0.255)
SeaTime_i				3.278 (2.312)
$\text{SeaTime}_i \times \text{Air}_i$				-3.993** (1.960)
Observations	8862	8857	8491	8369
t FE	✓	✓	✓	✓
Time to next shipment FE		✓	✓	✓
HS6 FE			✓	✓
Destination country FE			✓	
Gravity controls				✓

Note: This table displays the results from estimating equation (2). “Time to next shipment FE” is a fixed effect for the number of months between the export shipment and the next export shipment. Gravity controls include latitude and longitude of the destination country, a dummy for landlocked, contiguousness, past colonial experience and a continent fixed effect. All regressions are weighed by export value. Robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

¹²We drop the rare cases in which the firm exported by both modes in the same month. Appendix Table B.2 repeats the exercise replacing Air_i by the share of exports shipped by air. The results are almost identical.

Table 3 displays the results. Throughout all specifications 1 to 3, the coefficient on the air variable is negative and significant, implying that shipments sent by air are paid earlier than those shipped by sea, even controlling for shipment size, good, and destination. The coefficients imply that an export shipped by air results in an earlier receipt of the payment by 2 to 3.5 months.¹³ In column 4, we further control for the shipping time by sea from Colombia to the country of destination, also controlling for typical gravity variables.¹⁴ Payment lag is larger for destinations with longer shipping times, but not for air exports. We view this as suggestive evidence of our mechanism: longer shipping time leads to longer payment delay, consistent with (at least part of) the payment being made at time of delivery.¹⁵

The payment delay differential between modes, combined with the freight cost premium for air shipping, implies that air shipping can be used as a substitute for external finance in the presence of payment lags: by choosing air, firms accelerate cash collection and reduce their working-capital needs, at the cost of higher freight rates.¹⁶ The borrowing rate at which firms would be indifferent between the two modes can be inferred by combining the cost differential from table 2 with the payment-lag differential from table 3.¹⁷ It ranges between 18% and 37%, consistent with Colombian borrowing costs for firms in the upper end of the interest rate distribution, providing a quantitative sense of the financial friction at the heart of our mechanism. Importantly, the mechanism is not relevant for all firms equally — it binds most tightly for those facing the highest borrowing costs. Moreover, interest rates are only one dimension along which credit conditions vary; lenders also adjust loan quantities, maturities, and collateral requirements (Matray et al., 2024; Amiti and Weinstein, 2011; Feenstra et al., 2014). In the next section, we estimate the effect of broader measures of credit supply on the probability of shipping by air, capturing this full range of variation rather than the price margin alone.

Limitations. The exercise in this section has some limitations. First, the sample is restricted to the first export. Appendix B.1.2 repeats the exercise expanding the sample to any transaction occurring at least a year after the previous one, with similar results. In Appendix

¹³We note that this does not imply late payment. On the contrary we believe it reflects a different contractual payment agreement.

¹⁴We obtain this data by requesting quotes from three main shipping companies from Colombia to the destination and take the average number of days. The data is an imperfect measure of actual shipping time, since it ignores further inland transit.

¹⁵Although the main effect of SeaTime_i is not significant at standard levels, the sign is positive. Since SeaTime_i is measured with error, we don't view this as problematic especially since the magnitude is perfectly offset by the interaction with Air_i . Appendix Table B.4 presents results from unweighted regressions where the main effect is significant.

¹⁶We acknowledge that the two findings rely on two different datasets and samples. Appendix A.6 provides a reassuring description of the comparability between the two samples.

¹⁷See Appendix B.1.4 for details on the computation.

B.1.2, we show there is no difference in advance payment between air and sea, so there is no loss from focusing on payments happening after the shipment. Appendix B.1.3 provides an alternative event-study estimation strategy, which reveals that most air shipments are paid within a month while sea shipments take up to six months. Finally, we cannot see domestic payments that the firm could receive from selling its claim on the importer to a third party. Nevertheless, those factoring arrangements typically come at a discount, so that they also impose a “financing” cost to the firm.

Fact 2: Firms with higher access to finance use fewer air shipments

Now, we show that firms with more access to finance reduce the use of air mode. We start by estimating the following equation to capture the effect of higher access to banking credit on the probability of air shipments:

$$\text{Air}_{odgjmts} = \beta_0 + \beta_1 \text{creditAccess}_{jt} + \beta_2 \log(\text{unitValue})_{odgjmt} + \alpha_{ts} + \alpha_{odgm} + X'_{jt}\beta + \epsilon_{odgjmts}, \quad (3)$$

where $\text{Air}_{odgjmts}$ takes the value of one if the export transaction uses air, and zero otherwise, for a shipment of firm j , in month m , year-sector ts of good g from origin o to destination d . The coefficient of interest β_1 measures how firm’s access to credit, creditAccess_{jt} , affects the probability of shipping by air for those firms that have at least one loan.

We use four complementary measures of access to finance: Interest payments to sales, interest rate, total debt, and number of loans. Our preferred measure of access to finance is the ratio interest payments to sales, calculated as annual interest paid divided by initial sales. It measures the effective use of external financing, meaning that higher interest payments to sales indicates higher debt-dependence (Feenstra et al., 2014). For this ratio, we keep sales constant for every firm and use their first year sales t_1 to avoid capturing the dynamics of firms’ sales. The average interest rate represents the cost of capital: lower rates imply easier access, while higher rates reflect stricter and more expensive lending conditions. Total debt assesses the size of a firm’s bank borrowing, revealing its reliance on external finance regardless of the firm size. Lastly, we measure access in terms of the number of active loans. We construct a dummy for having a number of loans above the median, quantifying the breadth of a firm’s engagement with banks (Behr et al., 2011; Diamond, 1991); firms with more loans typically enjoy broader, diversified access to credit. Together, these measures allow us to assess how both the presence and scope of bank financing shape firms’ shipping choices. We standardize our first three measures. Hence, our results are interpreted as the semi-elasticities.¹⁸

¹⁸Notice that all our measures capture only the intensive margin of access to banking loans. We show in

We include a broad set of controls. First, we control for the unit value of each shipment, $\log(\text{unitValue})_{odgjt}$, as more valuable shipments are more likely to be shipped by air. Second, we include origin-destination-good-month fixed effects α_{odgt} , which capture specific modes for goods that are always traded between two cities during specific months of the year e.g., flower shipments from Bogotá to Mexico City via air in February. We consider HS6 codes for goods. We also include time-sector fixed effects, α_{ts} , to capture sectoral aggregate dynamics e.g., the global price of coffee.

Our identification compares firms in the same year and sector, shipping the same good at the same unit value, from the same origin to the same destination and month, by both transport modes. To control for firm observable characteristics that may independently drive mode choice — size, growth, financing sources, and liquidity — we include a lagged set of controls comprising sales, de-meaned leverage, inventory to assets, cash to assets, sales growth, and FDI receipt on the extensive margin (Foley and Manova, 2015; Evans and Harrigan, 2005; Deardorff, 2003; Iannotta and Kwan, 2022; Ottonello and Winberry, 2020; Hennessy et al., 2007).¹⁹ This between-firm variation captures most of the heterogeneity of interest: firms facing different credit conditions, at the same point in time, shipping the same product to the same market from the same city of origin. Column 4 of Appendix Table B.8 adds firm fixed effects, exploiting firms that ship the same good to the same destination and month by both modes; the estimates remain stable in sign and magnitude, confirming that the results are not driven by time-invariant firm-specific mode preferences.²⁰ Appendix Table B.10 re-estimates Equation (3) at the HS8 level; coefficients are nearly identical, confirming that finer product definitions do not affect the results.

Panel A of Table 4 shows that greater access to bank credit lowers the probability of shipping by air. A one-standard deviation increase in interest payments over sales—our main measure of banking access—reduces the air-share by about 3 percentage points (from 51% to 48%), or 5.7% of total air shipments. Alternative measures of credit access point in the same direction: higher interest rates are associated with greater air use, while higher total debt and a larger number of loans are associated with a lower air share. These alternative OLS coefficients are not statistically significant, however, which already points to the importance of isolating credit supply with an IV strategy.²¹ These results are consistent with better-

the appendix that the extensive margin is not a critical dimension of how access to credit affects the mode of transportation (see Table B.7 in the Appendix), where we regress the probability of shipping by air on a dummy variable equal to 1 if firm j has at least one active loan in year t , and interact each of the intensive margin measures with this dummy.

¹⁹See Appendix Section A for the construction of FDI receipt on the extensive margin.

²⁰Appendix Table B.9 also shows robustness to alternative fixed-effect combinations.

²¹Firm-level clustering is more conservative, as it allows for arbitrary correlation among a firm’s shipments across years. Appendix Table B.11 reports the same regressions clustering at the firm-year level, under which

Table 4: Higher Access to Banking Credit Reduces the Probability of Shipping by Air

	Prob. of shipping by air			
	(1)	(2)	(3)	(4)
Panel A: Baseline Estimates				
Int. payments to sales	-0.026** (0.013)			
Int. Rate		0.007 (0.005)		
Total debt			-0.011 (0.007)	
More loans				-0.011 (0.010)
Log(Unit value)	0.021*** (0.004)	0.020*** (0.004)	0.021*** (0.004)	0.021*** (0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
N	4,559,233	4,310,538	4,566,662	4,568,834
Panel B: IV Estimates				
Int. payments to sales	-0.288** (0.113)			
Int. Rate		0.075** (0.030)		
Total debt			-0.037*** (0.013)	
More loans				-0.257** (0.124)
Log(Unit value)	0.020*** (0.004)	0.020*** (0.004)	0.021*** (0.004)	0.019*** (0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
N	4,559,233	4,310,538	4,566,662	4,568,834

Note: This table displays the estimated coefficients for the impact of banking credit access on the likelihood of shipping by air. Panel A shows baseline estimates for equation 3 using alternative measures for β_1 : (1) Interest payments to sales (standardized); (2) Interest rate (standardized); (3) Total debt (standardized); and (4) Dummy for having a number of loans above the median. Panel B reports IV estimates, where the instrument is defined by the firm's exposure to healthier banks (see equation 4). The variable "Unit value" measures the shipment value divided by the number of units in logs. In both panels, total debt is total outstanding bank loans, interest payments to sales is annual interest payments to banks over first sales, and "more loans" is a dummy for firms above the sample median number of bank loans. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth and FDI receipt in the extensive, which are lagged by one year. Robust standard errors are clustered at the firm level. *** : $p < 0.01$, ** : $p < 0.05$, * : $p < 0.1$.

financed firms substituting away from fast, expensive air freight toward slower, cheaper sea freight, tolerating longer payment delays in exchange for higher profitability per shipment.²²

all four OLS coefficients are statistically significant. The IV estimates in Panel B are significant under both clustering choices.

²²Appendix Table B.12 shows similar signs for short- and long-term components of our intensive-margin credit measures.

Reverse causality and IV. One potential concern with our results is that they may be driven by changes in firms’ credit demand rather than by their access to credit. To address this issue, we isolate credit supply using an instrumental variable (IV) approach that follows the bank lending channel literature where we exploit firms’ pre-determined lending relationships to construct their differential exposure to bank-level supply shocks (Paravisini et al., 2015; Amiti and Weinstein, 2018).²³ Specifically, we exploit cross-bank variation in financial strength and loss-absorbing capacity, as measured by the Tier 1 capital ratio—a regulatory indicator defined as a bank’s core equity capital divided by its total risk-weighted assets (RWA). This ratio captures a bank’s capitalization relative to risk exposure, reflecting prudential and managerial choices rather than the borrowing, investment, or credit demand decisions of individual firms.

The channel through which this instrument operates is the well-documented link between bank capitalization and lending capacity: better-capitalized banks face fewer regulatory or funding constraints and hence can extend more credit (Correia et al., 2024; De Jonghe et al., 2020; Jiménez et al., 2017; Gropp et al., 2019; Ormazabal et al., 2025). The exogeneity assumption relies on the fact that Tier 1 capital ratios evolve primarily due to banks’ past profitability, capital issuance, and risk management policies—factors unrelated to contemporaneous firm-level shocks in credit demand.²⁴ The exclusion restriction further requires that bank capitalization affects firms’ air-shipping decisions only through this credit supply channel. This condition is supported by our setting: Colombian commercial banks are predominantly domestic institutions whose capitalization decisions are unlikely to operate through any channel other than domestic lending.²⁵

We therefore define our instrument, in the spirit of the firm–bank exposure designs of Khwaja and Mian (2008) and Amiti and Weinstein (2018), as the weighted average of a firm’s exposure to banks with stronger Tier 1 capital ratios, using lagged loan shares as weights to

²³Identifying the bank shock through within-firm comparisons across lenders, as in Khwaja and Mian (2008) and Amiti and Weinstein (2018), requires firms with multiple concurrent banking relationships—scarce once the sample is restricted to exporters. We instead follow the exposure-based construction of Paravisini et al. (2015), who face a similar constraint among Peruvian exporters, extended here to a setting where the bank-level shock varies continuously over the panel rather than around a single crisis event.

²⁴In the language of the bank lending channel and firm-bank exposure literature (Khwaja and Mian, 2008; Paravisini et al., 2015), the identifying assumption is that a firm’s lagged allocation of borrowing across banks is not systematically related to future firm-specific credit demand shocks, conditional on our firm-level controls and fixed effects; the instrument then isolates variation in credit conditions that firms inherit from their pre-existing banking relationships rather than variation they choose.

²⁵Notice that a natural alternative channel through which bank capitalization might affect trade is the provision of trade finance instruments such as letters of credit. However, in standard trade finance practice, letters of credit are opened by the importer at their own bank in favor of the exporter as beneficiary. The relevant issuing institutions are therefore the foreign buyers’ banks, entirely outside the Colombian banking relationships that constitute our instrument.

avoid contemporaneous sorting of firms across banks:

$$Z_{jt} = \sum_b \omega_{jbt-1} \times \text{Tier1}_{b,t} \quad (4)$$

where ω_{jbt-1} is the share of the average value of the loans that firm j had with bank b in period $t - 1$, and $\text{Tier1}_{b,t}$ is the tier I capital ratio of bank b on December of year t . An increase in Z_{jt} can be interpreted as an exogenous change in credit access for firm j . Using our instrument, we estimate Equation 3 using 2SLS. We provide additional discussion of the instrument’s relevance and exogeneity in Appendix B.2.2, and present the corresponding first-stage (with F-statistics above conventional thresholds), reduced-form, and instrument-balancing results in Tables B.14, B.15 and B.16, respectively. All results are consistent with our instrument constituting a credit supply shock: an increase in Z_{jt} leads to an increase in interest payments to sales, decrease in interest rate, increase in total debt and number of loans.

Panel B of Table 4 reports the second-stage results, showing that increased access to credit is associated with a lower probability of shipping by air, with all effects statistically significant at the 95% confidence level. The IV coefficient is about ten times larger than the OLS estimate. This can be explained by classical measurement error in each measure of credit access that biases the OLS coefficient towards zero. The instrument captures a broader credit supply shock, while each individual measure imperfectly captures access to finance. Following Hull (2025), we also show that the compliers are disproportionately large firms (see Table B.16). The larger IV coefficient can thus be explained if larger firms are also those with more capacity to adapt their mode of shipping. Far from being a limitation, this selection confirms that the instrument isolates variation precisely where the working capital channel operates most powerfully. For instance, Perez-Reyna et al. (2025) finds that larger and more productive firms in Colombia are those more affected by similar changes in credit supply.

A one-standard-deviation increase in interest payments to sales, exclusively driven by additional Tier 1 capital exposure, reduces the probability of shipping by air by 28 percentage points and lowers the share of air shipments from 51% to 23%, implying transportation cost savings of about 700 thousand dollars per firm a year or 1.7% of annual exports. In terms of CO_2 emissions, an average air transaction in our data weighs 1.4 tons, travels 4,000 km, and emits roughly 2.8 tons of CO_2 ; under the same shock, the resulting reduction in air shipments corresponds to an annual decrease of approximately 533,000 tons of CO_2 , or a reduction of 4.13% of total emissions in our sample. To put this scale in context, this reduction is equivalent to mitigating roughly 17.6% of Colombia’s total annual international aviation

CO₂ emissions, which stand at approximately 3.02 million tonnes ([Unidad Administrativa Especial de Aeronáutica Civil, 2024](#)).

The effects for interest rates and total debt are more modest but still sizable. A one percentage point increase in the interest rate raises the probability of shipping by air by about 7.5 percentage points, while a one-percentage point increase in the debt ratio lowers it by roughly 3.7 percentage points—non-negligible, but far below the 28-percentage-point effects estimated for interest-payments-to-sales and the number of loans. Because interest payments over sales scale debt service by firm size, they generate much larger effects on transport choice than the more unconditional debt ratio. This suggests that the cost of servicing debt, rather than leverage alone, is what primarily shifts firms away from air toward sea shipping.

Fact 3: Air intensity differs across goods

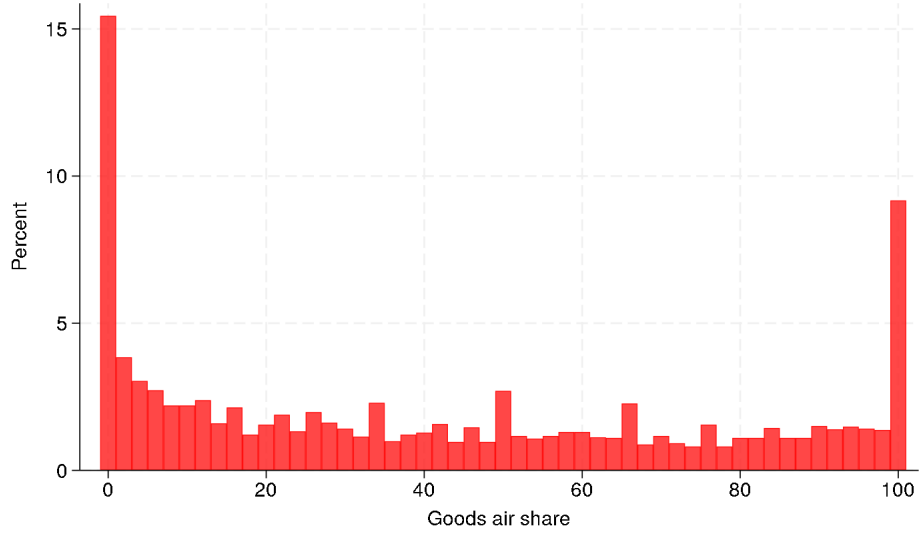
We document that air intensity varies by good, although a significant proportion of goods are shipped by both sea and air.²⁶ Figure 1 displays a histogram of the share of air shipment in HS6 codes, using all Colombian exports from 2008 to 2019 to compute the average use of air in all transactions. The horizontal axis shows the share of goods, and the vertical axis represents the frequency of air usage.

Around 15% of goods never use air shipments, and around 10% always use air. For around three-quarters of goods, shipments happen by both air and sea in Colombian exports. This fact mirrors the findings of [Tolva \(2024\)](#) and [Lugovskyy et al. \(2025\)](#). Appendix B.3.2 replicates this analysis while controlling for unit value and city of destination, and further explores the share of air shipments across different dimensions—including transactions, goods, value, CO2 emissions, and weight.

An important implication of Figure 1 is that the air usage intensity differs across different goods, ranging from 0 for bulky low-value items (e.g., wood in the rough) to 1 for high-value or fast-depreciating items that can’t survive the lengthy delay of sea shipping (e.g., cut flowers, blood prepared for therapeutic, prophylactic or diagnostic uses). Our findings are well documented (see eg [Hummels and Schaur, 2013](#)), and have implications for comparative advantage. If a country has a large cost of shipping by sea, it will specialize in air-intensive goods. Whether the country is completely precluded from exporting sea-intensive goods depends on the substitutability of modes within good.

²⁶We exclude from our sample all road exports for two reasons. First, Venezuela is a small trading partner in the sample period after years of negative growth. Second, the roads leading to the customs land ports on the Colombia-Ecuador border are infrequently used due to safety reasons.

Figure 1: Goods transport mode mix: 79% use both air and sea, 21% use only one



Note: The histogram displays the distribution of air intensity across goods exported in Colombia using HS6-digit classification. Air intensity is computed as the share of transactions within good that use air. Data Source: DIAN and author’s elaboration.

Implication of the three facts Our first two facts point towards a tradeoff between a more expensive air shipment in monetary terms, versus a more expensive sea shipment in terms of cost of credit. To ship a given good, firms can substitute credit with air shipments. Even with this margin of adjustment, however, the cost of shipping remains higher when the cost of credit is larger because of the higher air shipping cost. This is especially salient for goods that are sea-intensive. A higher cost of credit will therefore naturally lead to specialization towards air-intensive goods. Indeed, Appendix B.3.3 documents that countries with low financial development tend to export more air-intensive goods. In the next section, we build a model to quantify the relevance of this mechanism in shaping Colombia’s comparative advantage.

4 Model

In this section, we develop a theoretical framework to measure how better financial access affects a country’s comparative advantage through firms’ transportation choices, and calibrate it to the Colombian economy. Building on a two-country Krugman model with multiple sectors, working capital constraints, and endogenous mode choice, we distinguish two channels through which finance matters: financing time-to-build, highlighted in previous work, and financing payment lags. Through the lens of the model, the payment-lag channel accounts

for slightly over half of the total increase in exporting capacity, and 10 percent of the total impact of improved financial access on comparative advantage.

The economy consists of two regions, Colombia and the rest of the world (ROW), denoted by subscripts o or d . There are J sectors denoted by subscript j . In each region and sector, there is a continuum of firms that produce differentiated varieties. Each firm is denoted by subscript i . There is a different demand for goods shipped by air or sea (Hummels and Schaur, 2013). The population in any region o is given by L_o , and workers provide labor supply inelastically.

4.1 Demand

In each region d , there is a representative consumer with Cobb-Douglas preferences:

$$U_d = \prod_j (C_{dj})^{\alpha_{dj}} \quad (5)$$

where α_{dj} are the shares of each sector in total spending and

$$C_{dj} = \left(\int_{i \in \Omega_{dj}} a_{m(i)} (q_{di})^{\frac{\sigma}{1-\sigma}} \right)^{\frac{1-\sigma}{\sigma}} \quad (6)$$

where Ω_{dj} is the set of available goods within sector j in region d . The parameters $a_{m(i)}$ are taste shifters that depend on the mode of transportation of each good. In general, $a_{AIR} > a_{SEA}$ due to shorter delivery lags (Hummels and Schaur, 2013).

The expenditure in region d for the good of firm i shipped by mode m for a given price p_i is given by:

$$X_{i,odjm} = a_{mj} \left(\frac{p_{i,odjm}}{P_{dj}} \right)^{1-\sigma} \alpha_{dj} X_d, \quad (7)$$

where X_d is the aggregate expenditure and P_{dj} is the ideal price index of sector j in region d given by $P_{dj}^{1-\sigma} = \int_{i \in \Omega_{dj}} a_{m(i)} (p_{id})^{1-\sigma}$.

4.2 Supply

A firm i in region o , sector j , produces output using labor and a constant returns to scale production technology defined by $y_i = A_{oj} l_i$. The firm can hire labor at a wage rate w_o , and must pay it in advance i.e., the firm finances its (variable) wage bill before receiving the payment for its good. We assume that firms in region o , sector j , can borrow at a gross interest rate R_{oj} . In each sector, production takes $\bar{\tau}_j$ units of time. Therefore, absent

additional shipping times, the firm would have to pay $w_o (R_{oj})^{\bar{\tau}_j}$ for a unit of labor.²⁷

Firms use two modes to ship goods between regions, air and sea. The use of each mode leads to different trade costs. A firm in sector j and region o faces an average iceberg trade cost $d_{odjm} > 1$ to deliver goods to region d that depends on the shipping mode m . Thus, it needs to produce d_{odjm} units for one unit to reach the destination. A shipment from o to d by mode m takes τ_{odjm} units of time. Furthermore, the firm faces an idiosyncratic multiplicative shock $1/\varepsilon_{ijm}$ to ship by mode m . For a given transportation mode m , the firms' profits are given by:

$$\pi_{i,jm} = p_{i,jm} \frac{A_{oj} l_i}{d_{jm}} \varepsilon_{ijm} - w l_i (R_{oj})^{\bar{\tau}_j + \tau_{odjm}} \quad (8)$$

subject to the demand function in equation (7).

Firms implement a traditional constant mark-up pricing:

$$p_{i,jm} = \frac{\sigma}{\sigma - 1} \frac{w_o (R_{oj})^{\bar{\tau}_j + \tau_{odjm}} d_{odjm}}{A_{oj}} \frac{1}{\varepsilon_{ijm}}$$

and the mode-specific profits are given by

$$\pi_{i,jm} = \frac{1}{\sigma} \left(\frac{\sigma}{\sigma - 1} \frac{w_o (R_{oj})^{\bar{\tau}_j + \tau_{odjm}} d_{odjm}}{A_{oj}} \right)^{1-\sigma} (\varepsilon_{ijm})^{\sigma-1} a_{jm} \frac{\alpha_{dj} X_d}{P_{dj}^{1-\sigma}}.$$

The firm chooses the mode of transportation that maximizes its profits. Under the assumption that ε_{ijm} follows a Fréchet distribution with shape parameter μ , the probability that $\pi_{i,jm} \geq \pi_{i,jm'} \forall m'$ is given by:

$$s_{odjm} = \frac{\left((R_{oj})^{\tau_{odjm}} d_{odjm} (a_{jm})^{\frac{1}{1-\sigma}} \right)^{-\mu}}{\sum_{m'} \left((R_{oj})^{\tau_{odjm'}} d_{odjm'} (a_{jm'})^{\frac{1}{1-\sigma}} \right)^{-\mu}}. \quad (9)$$

Equation (9) implies that the probability that a firm uses a given mode depends negatively on the iceberg cost (d_{odjm}) and positively on the demand shifter for the mode (a_{jm}). Since the firm needs to finance its costs in advance and incurs an interest rate while waiting for the shipment to reach, it also favors a faster mode of transportation (with low τ_{odjm}) all else equal. Note that the time-to-build $\bar{\tau}_j$ itself doesn't affect the mode choice. This is

²⁷We introduce interest rate payments in this static model in a reduced form way, but our model can be shown to correspond to the steady state of a dynamic model. The gross interest rate R_{oj} captures in reduced form the financial friction mechanism in [Manova \(2013\)](#): higher borrowing costs raise firms' effective production costs, depressing exports from credit-sensitive sectors. We do not explicitly model credit markets, banks, or financial contracts; the empirical section provides direct evidence that the relevant variation in R_{oj} is credit supply driven.

because the trade cost is multiplicative, so that the cost of production is irrelevant to the mode choice. The model would still allow for any correlation between $\bar{\tau}_j$ and s_{odjm} as we keep d_{odjm} and a_{jm} as free parameters, but the impact of a change in R_{oj} on s_{odjm} , however, wouldn't be mediated by $\bar{\tau}_j$. If anything, ignoring this additional channel underestimates the importance of external finance on mode switching in our model.

The parameter μ represents the elasticity of substitution between air and sea mode for all sectors. There are two extreme cases that provide intuition behind this parameter. In the first case, $\mu \rightarrow \infty$ represents the case in which air and sea mode are perfect substitutes. Thus, firms use both modes to ship a good, firms never use a single mode to ship the same good, and firms split evenly the use of each mode for every good. In the second case, $\mu \rightarrow 0$, modes are not substitutable, thus, firms exclusively use a single mode for each good. That is, goods are either sea-shipping-only or air-shipping-only. For intermediate cases, firms can switch across mode, but there remain different overall air or sea intensities by sector depending on the values of exogenous parameters such as d_{odjm} and a_{jm} .

We define the *average* export cost d_{odj} by a weighted average of the mode-specific export costs, that is:

$$d_{odj} = \left(\sum_m \left[(R_{oj})^{\tau_{odjm}} d_{od,jm} (a_{jm})^{\frac{1}{1-\sigma}} \right]^{-\mu} \right)^{-\frac{1}{\mu}}. \quad (10)$$

Under the Fréchet assumption, total sales to destination d from origin o in sector j are given by:

$$X_{odj} = \gamma N_{oj} \left(\frac{\sigma}{\sigma - 1} \frac{w_o R_{oj}^{\bar{\tau}_j}}{A_{oj}} d_{odj} \right)^{1-\sigma} \frac{\alpha_{dj} X_d}{P_{dj}^{1-\sigma}} \quad (11)$$

where γ includes constants and the gamma function, while N_{oj} represents the mass of firms in region o , sector j .

Interest rate. There is a continuum of financiers in region o . They have deep pockets but need to hire labor to monitor credit. In particular, to monitor credit allocated for the payment of one unit of labor to sector j for τ periods, they need to hire $(R_{oj})^\tau - 1$ units of labor. Zero profit condition implies that they are willing to lend at a gross interest rate of

R_{oj} .²⁸ We assume that:

$$R_{oj} = (R^* + \bar{R}_o) \times \rho_j$$

where R^* , $\bar{R}_o$ and ρ_j are determined outside the model. $\bar{R}_o$ is a country-level wedge that depends on the country's level of financial development or the nation's financial misallocation. ρ_j is a sector-specific wedge, that can be rationalized by the presence of sector-specific monitoring costs.

Firms' entry. Firms in region o and sector j can enter the market by paying a fixed entry cost f_{oj} measured in units of labor. They pay the fixed cost before knowing their idiosyncratic ε_{ijm} shocks and interest rate. Firms keep entering until their expected profits are null.

Given the sales above, the zero-profit condition implies that variable profits are equal to the entry cost:

$$w_o f_{oj} N_{oj} = \frac{1}{\sigma} \sum_d X_{odj},$$

where we assume that the fixed cost can be paid at the end of the period so it doesn't require financing.²⁹

4.3 Equilibrium

Under balanced trade, total expenditure by consumers in region d is equal to total wage income: $X_d = w_d L_d$.

We formulate the equilibrium as a set of wages $\{w_d\}$, trade flows $\{X_{odj}\}$, the number of firms per region $\{N_{oj}\}$, and prices $\{P_{dj}\}$ subject to:

- Gravity equation

$$X_{odj} = \gamma N_{oj} \left(\frac{\sigma}{\sigma - 1} \frac{w_o (R_{oj})^{\bar{\tau}_j}}{A_{oj}} d_{odj} \right)^{1-\sigma} \frac{\alpha_{dj} w_d L_d}{P_{dj}^{1-\sigma}}$$

²⁸An other way to micro-found this setting is to say that the firms outsource labor payment entirely to the financiers. The financiers need to hire $(R_{oj})^\tau > 1$ units of labor to supply one unit of labor to sector j because some labor is required to check the firm's background, monitor that the workers are indeed used for production, and monitor the firm until payment is received after τ periods. Thus the firm pays $w_o (R_{oj})^\tau L_{oj}$ to the financier, and the financier hires $(R_{oj})^\tau L_{oj}$ units that it pays w_o . R_{oj} thus essentially acts as a wedge on labor productivity.

²⁹This model is equivalent to the steady state of a dynamic model, where the firm can use receipt of past sales to pay a per-period entry cost.

- Sectoral price index for every location

$$P_{dj}^{1-\sigma} = \gamma \sum_o N_{oj} \left(\frac{\sigma}{\sigma-1} \frac{w_o (R_{oj})^{\bar{\tau}_j}}{A_{oj}} d_{odj} \right)^{1-\sigma}$$

- Zero-profit condition

$$w_o f_{oj} N_{oj} = \frac{1}{\sigma} \sum_d X_{odj}$$

- Balanced trade

$$w_o L_o = \sum_j \sum_d X_{odj}$$

The expression for sectoral trade costs or *average export cost* between locations o and d , d_{odj} , is given by equation (10).

4.3.1 Equilibrium in changes

To reduce the number of parameters that we need to calibrate or estimate, such as productivity A_{oj} or mode-specific trade costs d_{odjm} , we reformulate the equilibrium conditions in changes as in Dekle et al. (2007). We denote proportional changes between the new and the old equilibrium as $\hat{X} = \frac{X^{new}}{X^{old}}$. Thus, we can solve for changes in average trade costs $\hat{d}_{odj}$, trade flows $\hat{X}_{odj}$, prices $\hat{P}_{dj}$, wages $\hat{w}_o$ and firm entry $\hat{N}_{oj}$:

$$\hat{d}_{odj} = \left(\sum_m \left((\hat{R}_{oj})^{\tau_{odjm}} \hat{d}_{od,jm} \right)^{-\mu} s_{od,jm} \right)^{-\frac{1}{\mu}} \quad (12)$$

$$\hat{X}_{odj} = \hat{N}_{oj} \left(\frac{\hat{w}_o (\hat{R}_{oj})^{\bar{\tau}_j}}{\hat{A}_{oj}} \hat{d}_{odj} \right)^{1-\sigma} \frac{\hat{w}_d \hat{L}_d}{\hat{P}_{dj}^{1-\sigma}} \quad (13)$$

$$\hat{P}_{dj}^{1-\sigma} = \sum_o \pi_{odj} \hat{N}_{oj} \left(\frac{\hat{w}_o (\hat{R}_{oj})^{\bar{\tau}_j}}{\hat{A}_{oj}} \hat{d}_{odj} \right)^{1-\sigma} \quad (14)$$

$$\hat{w}_o \hat{f}_{oj} \hat{N}_{oj} = \sum_d \frac{X_{odj}}{X_{oj}} \hat{X}_{odj} \quad (15)$$

$$\hat{w}_o \hat{L}_o = \sum_j \sum_d \frac{\alpha_{dj} \pi_{odj} X_d}{X_o} \hat{X}_{odj} \quad (16)$$

where π_{odj} is the initial equilibrium's trade share $X_{odj} / \sum_n X_{ndj}$. Equations (12) to (16) allow us to solve for the change in equilibrium based on data on initial modal shares ($s_{od,jm}$), trade shares (π_{odj}), sectoral shares ($\frac{X_{odj}}{w_o L_o}$) and destination output shares ($\frac{X_{odj}}{X_{oj}}$), as well as initial gross interest rates R_{oj} and elasticities $\sigma - 1$ and μ .

Two channels of financial cost We note that when holding d_{odj} constant in the gravity and sector price index equations (13) and (14), the term $\left(\hat{R}_{oj}\right)^{\bar{\tau}_j}$ is isomorphic to a change in sectoral productivity $\hat{A}_{oj}$. Smaller cost of finance is thus akin to productivity gain for sectors with large time to build $\bar{\tau}_j$. This channel captures mechanisms that previous literature has highlighted. Better access to finance drives comparative advantage towards finance-intensive sectors (Kletzer and Bardhan, 1987; Manova, 2013).

External financing costs also enter equilibrium equations through trade costs d_{odj} , but does so differently depending on exposure to mode of transportation. This is the novel channel our paper seeks to isolate. Equation (12) implies that the elasticity of average trade cost to a change in interest rate R_{oj} is given by:

$$\frac{\partial \ln d_{odj}}{\partial \ln R_{oj}} = \sum_m s_{od,jm} \tau_{odjm}. \quad (17)$$

If shipping time is ignored ($\tau_{odjm} = 0$), a change in interest rate has no effect on trade costs. If all modes of transportation had the same shipping lag ($\tau_{odjm} = \tau_{odj}$), then the change in trade cost would be unrelated to mode-specificity. This could accommodate settings in which shipping costs are unrelated to delivery times but still require financing.³⁰ However, when air shipments receive a faster payment than sea shipments as evidenced above, sea-intensive sectors are more sensitive to a change in interest rates. We note that the elasticity of substitution μ doesn't matter to a first order for the change in trade costs.

Accordingly, in our counterfactual exercises below, we will decompose the impact of changes in R_{oj} into a “productivity channel” where we leave trade costs unaffected and a “trade cost channel” where we shut down the change in R_{oj} while exogenously changing d_{odj} by the amount implied by the decrease in R_{oj} .

³⁰For example, in Manova (2013), a fraction of a fixed cost of exporting must be paid in advance and necessitates credit. We ignore this channel in our calibration since setting $\tau_{odj,AIR} = 0$ guarantees that there is no particular financing required to export outside of the shipping lag in sea. Our results on the impact of financial access through trade costs can thus be viewed as a lower bound.

4.4 Calibration

In addition to sectoral and trade shares, the following parameters have to be calibrated: the trade elasticity ($\sigma - 1$), the sector-specific time-to-build ($\bar{\tau}_j$), the initial gross interest rate (R_j) and the shipping mode elasticity μ .

We use estimates of the trade elasticity from the literature, and calibrate sectoral and trade shares directly from the data. For interest rates, we use the US real interest rate 2012-19 average for $R^* = 1.02$, and set the rest of the world's premium ($\bar{R}_{ROW}$) to 0, while we set $\bar{R}_{COL}$ to match the average Colombian real interest rate of 1.08. Sectoral rate premiums are set based on the observed average rates in the *Super Sociedades* data.

We calibrate the sector-specific time-to-build ($\bar{\tau}_j$) to match the aggregate short-term debt interest payments to total variable cost in the *Super Sociedades* data. Appendix C.1 describes the exact procedure and compares our sectoral estimates to existing measures.

Our sectoral breakdown corresponds to slight aggregates of the ISIC 2-digit classification, so that the model is composed of two regions (Colombia and a rest of the world composite), 32 sectors (mostly manufacturing) and 2 modes of transportation (sea and other). In calibrating shipping times, we interpret mode-specific delivery lags as cash-recovery lags rather than pure physical transit times. Table 5 describes the source of our calibration. The only parameter remaining to estimate is the shipping mode elasticity that we estimate below.

Table 5: Calibration summary

Object	Related to	Value	Source
$\pi_{odj}, \frac{X_{odj}}{w_o L_o}, \frac{X_{odj}}{X_{oj}}$	Trade and sectoral shares		ICIO (OCED)
$s_{od,jm}$	Modal share		Customs data
$\bar{\tau}_j$	Time-to-build (in years)	[0, 1.09]	Supersociedades
R^*	World interest rate	1.02	World Bank (US)
$\bar{R}_o$	Interest premium	$\bar{R}_{ROW} = 0, \bar{R}_{Col} = 0.0612$	World Bank
ρ_j	Sectoral rate	[-0.017, 0.045]	Supersociedades
σ	Trade elasticity - 1	4	Literature
τ_{odjm}	Shipping time	$\tau_{SEA} = 3.5/12, \tau_{AIR} = 0$	Table 3
μ	Mode elasticity	4.8	Own estimate

Note: The table displays the calibrated values for parameters and the data sources. We report the range of sectoral values for $\bar{\tau}_j$, see Appendix C.1 for details on the calibration. The table reports the range of ρ_j which is computed as the deviation of sectoral interest rate from the average rate.

4.4.1 Shipping mode elasticity

To estimate μ , we start from equation (9) on the modal share reproduced below for convenience, adding the time dimension we also have in the data:

$$s_{odjmt} = \frac{\left((R_{oj,t})^{\tau_{odjm}} d_{odjm} (a_{jm,t})^{\frac{1}{1-\sigma}} \right)^{-\mu}}{(d_{odj,t})^{-\mu}}$$

$$s_{odjm,t} = \exp \left(-\mu \times \tau_{odjm} \ln R_{oj,t} - \mu \ln d_{odjm} + \frac{\mu}{\sigma - 1} \ln a_{jm,t} + \Phi_{odj,t} \right)$$

Focusing on Colombian export data, we drop the o subscript and parametrize the trade cost and taste shifter as a djm fixed effect to get the following estimation equation

$$s_{odjm,t} = \exp \left(-\mu \times \tau_{djm} \ln R_{oj,t} + \eta_{djm} + \gamma_{djt} \right). \quad (18)$$

The remaining hurdle is to specify τ_{djm} , the amount of time before the firm receives payment for a shipment by mode m . Since our model is calibrated to Colombia and the rest of the world, we simply use a dummy variable for sea shipments, and interpret the magnitude based on the delay in payment documented above.

Table 6: Mode switching elasticity estimation

	Modal share ($s_{Col,djmt}$)			
$1_{m=AIR} \times \ln R_{ojt}$	1.286** (0.443)	1.431** (0.583)		
$\text{time}_{Col,md} \times \ln R_{ojt}$			-24.55*** (10.07)	-35.83* (20.57)
Observations	155722	144982	144992	129844
Control for $1_{m=AIR} \times \ln \frac{\text{value}}{\text{weight}}$	✓	✓	✓	✓
$odjt$ and $odjm$ FE	✓	✓	✓	✓
$odmt$ FE		✓	✓	✓
$ojmt$ FE				✓

Note: Results from estimating equation (18) with PPML. $\text{time}_{Col,d}$ is set to 1/365 for air shipments and $n_d/365$ for sea shipments, where n_d is the number of days take for a ship to reach d from Colombia. Standard errors are clustered at the sector-mode-time level.

Table 6 shows the results of estimating equation (18) using a Poisson PML estimation. The first two columns simply use a dummy variable for sea instead of τ_{odjm} , so that the coefficient should be interpreted as the elasticity of mode-switching $-\mu$ multiplied by the

average time difference when receiving payment for sea versus other shipments. Hence, the coefficient of 1.4 in the second column implies an elasticity $\mu = 1.4/(3.5/12) \approx 4.8$ given that sea exports are paid about 3.5 months after air exports (Table 3). The estimate in column 2 also includes a destination-mode-time fixed effect that could capture any time varying demand shift for a particular mode or any overall change in shipping cost for that mode.

Estimates of the elasticity of substitution between modes are scarce. First, Fuchs and Wong (2024) obtain an elasticity estimate of around 1 for rail and truck modes in the U.S. Second, Tolva (2024) estimates an elasticity of around 2.6 for air and sea. Our estimate is higher, and section 5.2 explores the sensitivity of our model results to the value of μ .

As a robustness check, in columns 3 and 4, we instead measure τ_{odjm} as the number of days ships take from Colombia to destination d 's main port (or closest port for inland destination). As expected, the coefficient is negative and significant, indicating that shipments by sea are less likely when interest rates are higher, especially for destinations that take more time to reach. Column 4 even includes a sector-mode-time fixed effect, thus only using variation in time to ship across destination.³¹

5 Counterfactual: Improving access to finance

We use the model to study the impact of financial development. We remove the financing wedges faced by Colombia relative to the rest of the world by exogenously reducing R_{oj} for all sectors in Colombia. We decompose the results into the two channels outlined above: a “trade cost channel” that isolates the impact of R_{oj} on trade costs and a “productivity channel” that shuts down the change in trade costs.

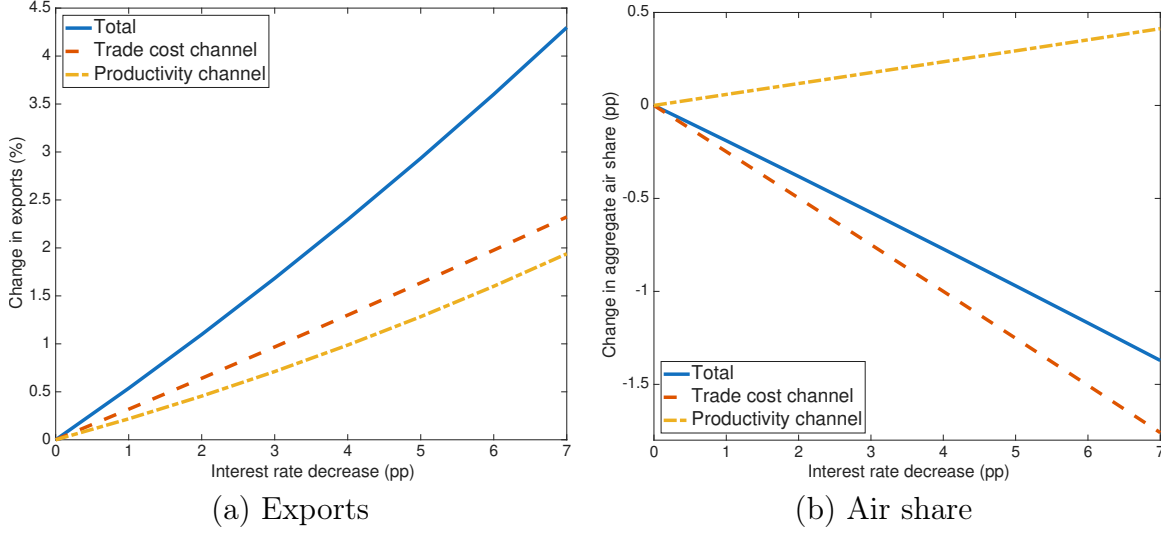
Figure 2 displays the impact of a variety of interest rate decreases on Colombia’s aggregate exports (left panel) and on modal share (right panel).

The left panel shows that when interest rate decreases by 7 pp (about the gap between Colombia and US interest rates), Colombian exports grow by above 4%. Slightly more than half of this is driven by the decrease in export costs due to lower interest payment during transit (dashed red line). The rest is due to the productivity channel (dotted yellow line) where overall production costs decrease due to lower interest payment during the time to build.

The right panel depicts the change in aggregate air share of Colombian exports. The vertical axis depicts the change in aggregate air share for each decrease in interest rate (solid

³¹If our measured shipment time was equal to the time delay in receiving payment, one could interpret the coefficients directly as the mode-switching elasticity. However, we refrain from doing so as the time of shipment is measured with error and we can instead rely on the direct evidence on average delay in payments received by mode.

Figure 2: Aggregate exports and air share when interest rate decreases



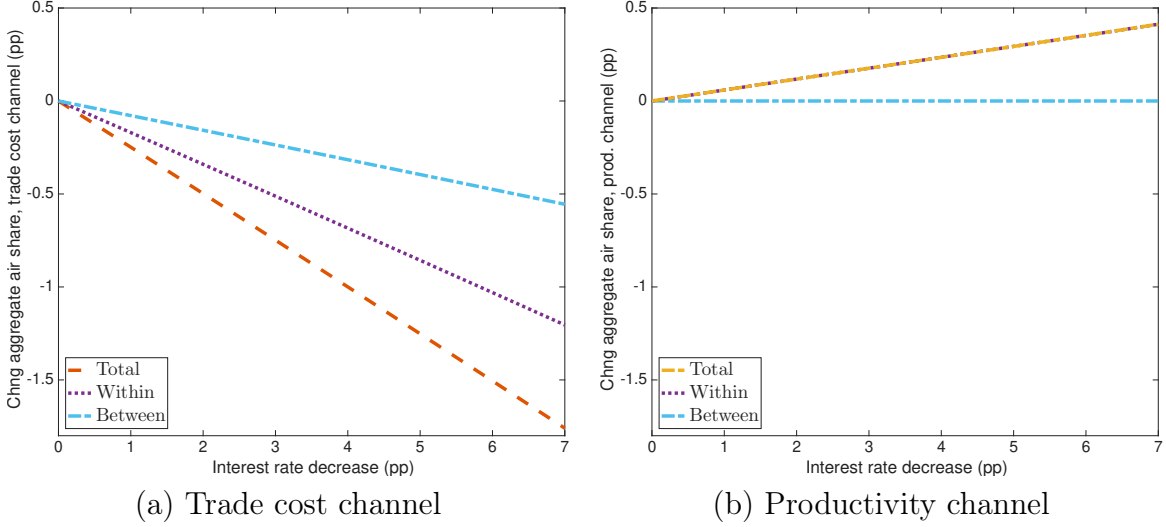
Note: In this figure, we display the results of multiple counterfactual scenarios in which we reduce the interest rate faced by all Colombian sectors. For each counterfactual, each point represents the change in aggregate exports or air share of Colombia's exports (y-axis) when we reduce the interest rate by a specific amount measured in percentage points (x-axis).

blue line). When interest rate decreases by 7 pp, the share of air shipments decreases by around 1.4 pp, from a base share of 27% in value. The trade cost channel leads to a decrease in air share. This is due to two effects. First, within a sector, firms substitute towards sea shipping because the delivery lag becomes less costly in terms of interest payments. Second, trade costs decrease by more in sectors that are sea-intensive, so that their importance in aggregate exports increases. The trade cost channel is thus unambiguously negative. On the other hand, the productivity channel can have either impact on aggregate air share. The decrease in interest rate lowers the financing cost borne by sectors with long time-to-build $\bar{\tau}_j$, thereby shifting the composition of exports towards sectors with high values of $\bar{\tau}_j$. Whether this increases or decreases aggregate air share depends on the correlation between $\bar{\tau}_j$ and air-intensity. The dot-dash yellow line plots the productivity channel and shows that it has a positive impact on air share, indicating that there is a positive correlation between $\bar{\tau}_j$ and air-intensity, a fact we return to below.

To further investigate the importance of sectoral reallocation in driving the change in air share, we decompose the change into a within-sector change and a between sector reallocation change defined as follows:

$$s_1^{AIR} - s_0^{AIR} = \sum_j \underbrace{\frac{\lambda_j^0 + \lambda_j^1}{2}}_{\text{export share}_j} \underbrace{(s_{odj,AIR}^1 - s_{odj,AIR}^0)}_{\text{change within sector}} + \sum_j \underbrace{(\lambda_j^1 - \lambda_j^0)}_{\text{between/reallocation}} \underbrace{\frac{s_{odj,AIR}^1 + s_{odj,AIR}^0}{2}}_{\text{avg air share}_j}$$

Figure 3: Within and between sector reallocation and air share



Note: The left panel decomposes the trade cost channel into within and between effect. The dashed red line is identical to that of the right panel of Figure 2. The right panel decomposes the productivity channel into within and between effect. The dash-dot yellow line is identical to that of the right panel of Figure 2.

Figure 3 plots the within-between decomposition in the two cases, in addition to the original decomposition. The left panel plots the impact of the trade cost channel. Both the within (dashed red) and between (dash-dot yellow) lines are negative, with the within-sector accounting for about two-third of the total change in air share. The right panel plots the impact of the productivity channel. The within term is null, since we shut down changes in trade costs, but the between channel is positive, implying that sectors with larger $\bar{\tau}_j$ are actually more air-intensive. This is consistent with [Harrigan \(2010\)](#) evidence that lighter, higher-value goods are disproportionately shipped by air and with [Hummels and Schaur \(2013\)](#) finding that sectors with high time costs rely more on fast air transport.³² Appendix Figure C.3 shows that the correlation between $\bar{\tau}_j$ and air share is indeed positive.

5.1 Drivers of comparative advantage

The results above imply that the two channels have an opposite impact on sectoral reallocation. The trade cost channel shifts sectoral allocation towards sea-intensive sectors, while the productivity channel shifts towards air-intensive sectors. The impact on aggregate air share is dominated by the fact that sea-intensive sectors expand, but the impact on overall sectoral composition of exports can go the other way.

³²In our model, there is no mechanical relationship between $\bar{\tau}_j$ and air intensity because the financing cost for production operates regardless of the mode of shipment. From the point of view of the model, this positive correlation between $\bar{\tau}_j$ and air share can be rationalized by a positive correlation between $\bar{\tau}_j$ and the sector-mode specific shipping costs $d_{odj,sea}/d_{odj,air}$ or sector-mode specific taste shifters $a_{j,air}/a_{j,sea}$.

To uncover the relative importance of drivers of the overall shift in sectoral composition of Colombia’s exports in the counterfactual, we compute the change in the Balassa Index of Relative Comparative Advantage for each sector when the aggregate interest rate in Colombia decreases by 7%. We then regress it on the two main sources of sectoral heterogeneity: baseline sea intensity and overall sectoral exposure to finance through the time-to-build.

Table 7 displays the results of these regressions. The first column shows that there is no unconditional correlation between change in RCA and sea share. On the contrary, there is a positive and significant correlation between change in RCA and exposure to finance (column 2). However, this doesn’t mean that the sea intensity doesn’t matter for the evolution of comparative advantage. When including both regressors, the correlation with initial sea share becomes significant and positive, and the correlation with $\bar{\tau}_j$ is strengthened. As discussed above, high-finance dependent sector (those with large $\bar{\tau}_j$) tend to ship more by air so that column two suffers from omitted variable bias.

In terms of magnitude, the time-to-build sectoral heterogeneity explains the larger share of the variance in revealed comparative advantage. It can account for 76% of the variance in RCA by itself, while adding the sea-intensity channel improves this to 83%, implying a partial R-square of 7%. Overall, this suggests that the importance of the sea-intensity channel is still sizable, at about 10% of that of the well-established channel of finance exposure.

Table 7: Channels of change in comparative advantage when decreasing interest rate

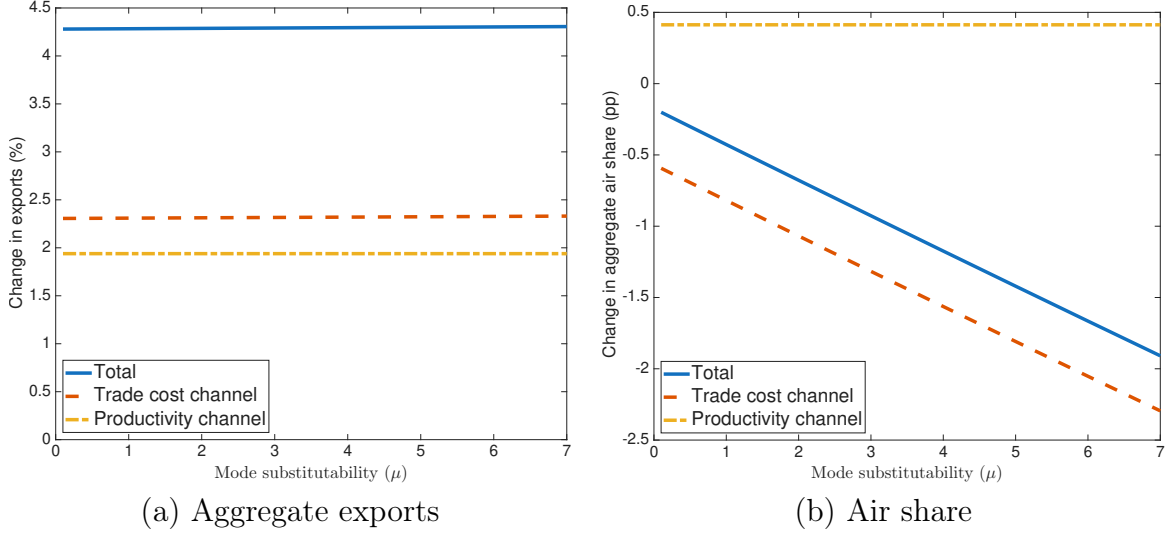
	Change in RCA		
Initial sea share ($s_{Col,ROW,j,sea}$)	-0.025 (0.037)		0.060*** (0.017)
Time-to-build ($\bar{\tau}_j$)		0.202*** (0.021)	0.230*** (0.019)
R2	0.015	0.762	0.832
N	32	32	32

Note: The table reports the results from regressing the change in sectoral RCA on the initial sea share and the time-to-build parameter ($\bar{\tau}_j$). All regressions are weighted by initial export value.

5.2 Sensitivity to the elasticity of substitution between modes

One of the key parameters of the model is the mode-switching elasticity μ . That parameter governs to a first order how much the air share within a sector reacts to a change in interest rate. However, μ does not matter for the change in trade cost to a first order (equation 17).

Figure 4: Relevance of mode switching elasticity



Note: The figure shows the impact of reducing the aggregate Colombian interest rate from 8.2% to 2% for different values of μ . The left panel depicts the impact on aggregate exports and the right panel depicts the impact on air share.

Hence, aggregate export changes are largely unaffected by the value of μ as shown in the left panel of Figure 4.

The right panel of Figure 4 illustrates the importance of the mode switching elasticity for air share, by plotting the change in aggregate air share for different values of μ . The within-sector channel is increasing in μ while the sectoral reallocation (“between”) channel is largely unaffected and would operate even in the absence of any switching, since it is driven mostly by the relative change in export costs and by sectoral time-to-build.

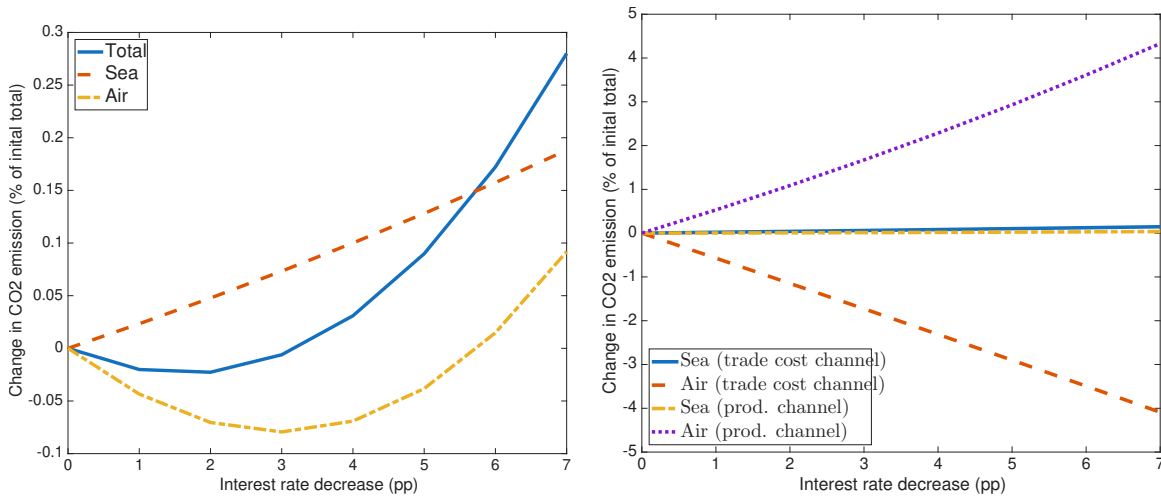
5.3 Environmental implications

As emphasized in [Lugovskyy et al. \(2025\)](#), the substitutability between air and sea shipping is relevant for the environment since air shipping is more CO_2 intensive. Our findings imply that a fall in interest rate could reduce air shipping and increase sea shipping, thereby decreasing overall CO_2 emissions. On the other hand, the sectoral reallocation channel is ambiguous as the previous section implies a potentially large reallocation towards air-intensive sectors due to the positive correlation between air-intensity and finance-dependence.

We investigate this in our model and compute the change in CO_2 emissions from Colombian exports caused by a decrease in interest rates, assuming that ocean transportation emits 103.5 times less CO_2 than air shipping ([Shapiro, 2016](#); [Lugovskyy et al., 2025](#)). Figure 5 plots the change, decomposed into emissions from air and sea shipment. The magnitude of the change is small, at around 0.2%. Overall sea emissions increase, as exports increase and

firms shifts towards sea. The impact on air emissions is U-shaped. At first, they decrease as air shipments are replaced by sea shipments. However, as sectoral reallocation towards high time-to-build sectors occurs, air emissions increase because those sectors tend to be air-intensive. The right panel illustrates this point. When shutting down the shift towards high time-to-build sectors (dashed red line), emissions would drop by an order of magnitude more. Note that even if at the 7% interest rate decrease, the net impact on emission is null, exports themselves are increasing thanks to the decrease in export cost and the decrease in production cost, so that emission per export are still decreasing.³³

Figure 5: CO2 emissions and interest rate decrease



Note: The left figure shows the estimated change in CO2 emissions following a decrease in Colombian interest rate, decomposed into the change coming for air and sea shipments. The right panel further decompose the change in air emissions isolating the reallocation towards sea-intensive sectors (red dash) vs high time-to-build sectors (dotted purple).

6 Conclusion

In this paper, we show that transport-mode-specific payment lags are a quantitatively relevant determinant of how access to finance shapes export patterns. Using Colombian data that link banks, firms, exports, and export payments, we document that sea shipments entail substantially longer payment lags than air shipments and that better access to bank credit tilts firms away from air freight and toward sea. We also document that improving access to finance reduces the likelihood of shipping by air.

³³We note that this quantitative results depends on the calibration. Notably, when the mode-switching elasticity is sufficiently low or the trade elasticity is sufficiently high, the result will not hold because trade flows increase by more than what the switching can compensate for.

Using a two-country model of international trade with external financing and transportation choices, we show that mode-specific payment lags shape both exporting capacity and sectoral specialization. Sea-shipping payment lags alone drive over 50% of the exporting-capacity gains from improved financial access. Moreover, these sea-shipping payment lags also account for about 10 percent of the sectoral-specialization gains from improved financial access.

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Online Appendix

Appendix A Data and Institutional Background

A.1 Institutional Background

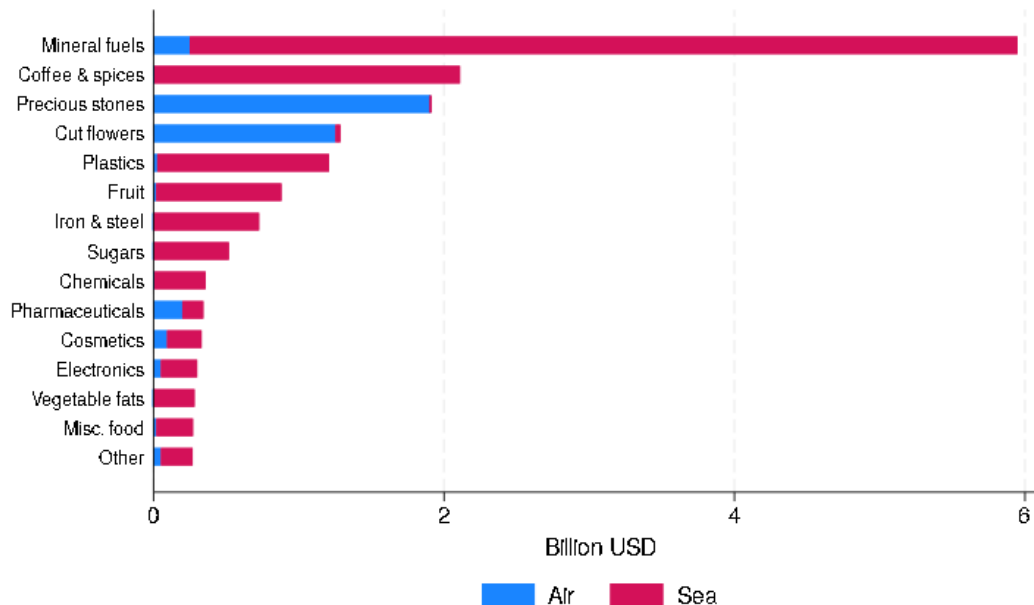
This section provides context on Colombia’s export structure, financial system, and its position in the cross-country distribution of air shipping and financial development.

Export structure. Figure A.1 shows average annual exports by HS2 sector over 2008–2019, split by transport mode. Colombia’s export basket is dominated by mineral fuels, which account for the vast majority of export value and are shipped almost entirely by sea. Beyond minerals, the main export sectors are coffee and spices, precious stones, cut flowers, and plastics. A striking feature of the figure is that virtually every sector uses both modes, though with clear specialization: cut flowers and coffee are predominantly air, while bulk commodities such as iron and steel, sugars, and fruit are predominantly sea. Notably, even mineral fuels exhibit some air use.³⁴ Importantly, the composition of Colombia’s export basket — raw materials, agricultural commodities, and intermediate inputs — suggests that Colombian exporters are predominantly selling to large buyers abroad for further processing or resale, rather than shipping directly to final consumers. This makes Colombia a natural small open economy setting where exporters are likely price-takers in both goods and shipping markets.

Financial system. Colombia’s financial sector is small relative to advanced economies and dominated by domestic banks, with bank credit representing the primary source of external financing for firms. Lending is concentrated among three institutions — Bancolombia, Banco de Bogotá, and Davivienda — which together account for 58.5% of corporate loans, while no single bank is completely dominant. Banks operate across all major cities and sectors without systematic specialization, and the equity market is thin: fewer than 200 firms are publicly listed, most of them financial. Firms therefore rely heavily on bank credit to finance operations, making the cost of credit a first-order determinant of firm behavior. For a detailed description of the Colombian financial system, see [Aristizábal-Ramírez and Posso \(2021\)](#). Corporate borrowing rates are substantially higher than in advanced economies — the composite cost of borrowing for euro area corporations averaged approximately 1.5% over 2008–2019 ([European Central Bank, 2019](#)) — and broadly in line with the Latin American average ([World Bank, 2024](#)). Figure A.2b shows the distribution of interest rates across firms in our three samples. Borrowing rates are concentrated between 3 and 10 percent, but the distribution has a long right tail, with a non-trivial share of firms facing rates above 20 percent.

³⁴We exclude precious stones from our analysis, as this sector consists of extremely high-value, low-weight items whose mode choice is unlikely to reflect the working-capital mechanism we study.

Figure A.1: Annual Colombian Exports by Sector and Transport Mode



Note: The figure displays average annual export value by HS2 sector, in billions of USD, for the top 14 sectors by export value. All remaining sectors are grouped as Other. Export value is split by mode of transport (air and sea). Road exports and transactions without city information are excluded. Source: DIAN, 2008–2019.

A.2 Customs Data: Exports

All export shipments must be reported to the Colombian Tax and Customs National Authority (DIAN). This report contains detailed information on the shipment and the exporter at the transaction level, including data on the good, weight, mode of transportation, firms and cities of origin and destination, and the value of export. We use data issued between January 2008 and December 2019 for firms with a valid firm identifier, where we define an incorrect identifier for having fewer than six digits or being non numeric.

We further clean city names in order to accurately assign geographic coordinates and standardize city information. To do this, we eliminate non-alphabetic characters and common non-city terms, followed by a harmonization against a global reference lists of recognized cities. We apply deduplication and string normalization methods, including both direct and fuzzy matching approaches, to link export city-country pairs to the international cities dataset. For cases where automated matching was insufficient, a combination of country-specific rules and manual assignments ensured that all cities were accurately matched to their standardized forms.

To organize the transaction-level data in a monthly panel, we remove observations for which the destination country is Colombia or a free trade zone. Since there could be modifications, addendum, or anticipated declarations, we only keep final ordinary exports and

the first time of declaration. Finally, we have a monthly exports panel that contains the number of units shipped, export value, packing, and weight, in addition to the transport mode, payment method, origin and destination locations, customs information, and product type, defined as the mode within a month.

A.3 Exports' payments

All foreign exchange declarations for export transactions are reported to Colombia's Central Bank (Banco de la República) in format 2 of the international transactions data. This report contains information on the exporter and the exchange operation, including each payment that an exporter firm received, and the bank through which the transaction was realized. The data contains export transaction payments including foreign and local currency payments, credit card and payment facilitator transfers, payments from major commodities like coffee and petroleum, duty-free zone payments, trade credit, advance payments, expenses, in-kind settlements, commissions, deductions, and devolutions. We use the data reported between January 2008 and December 2022.

We created a monthly panel using this data. To this end, we keep the current information of the exchange declarations, which reflects the last update on the transaction, since there could be modifications or withdrawals. Additionally, we keep only the first time of declaration and remove negative values from the reported cash payments. As with the rest of the data sources, we keep only the information for firms with the Colombian firm identifier "NIT" and, in this case, we keep the firms that engaged in trade. To create the monthly panel, we summed each type of payment by firm and generated the bank and sector's mode.

Finally we merged monthly panel data on export payments with the customs monthly panel. With this dataset, we are able to identify the first and second instances in which each firm exported. Next, we balance the panel and subject it to a series of filters: (1) Firms with total exports or total payments equal to zero are excluded. (2) Trade credit and main payments recorded prior to the first export are replaced with zero, as such payments should, by design, occur after the export event. (3) Firms with recorded payments one year prior to the first export are dropped, as these likely reflect misreporting. (4) Payments are retained within a window of six months before and twenty-four months after the first export. (5) Firms with exclusively advance payments are excluded. (6) Firms for which received payments exceed the value of the export at the time of first export are removed. (7) Firms that exported only once are excluded due to potential outlier behavior. For the event study, we construct two payment measures from the data: total payments, defined as the sum of anticipated, short-term, and long-term payments; and main payments, which comprise only short-term payments.

A.4 FDI inflows

We obtain data on FDI inflows from *Formato 4* (*Banco de la República de Colombia*), which records the sender and receiver firms, transaction values, and types of investment. We use

this data to build a dummy variable indicating whether firm i received any FDI inflow on year t , which we include as control in Table 4.

A.5 Payment gap by mode of transport

To estimate the differential payment lag between air and sea shipments reported in Table 3, we build a balanced firm-month panel that links each firm’s export shipments to the foreign-currency inflows it receives. To construct it, we first collapse the customs data to the firm-month-mode level, summing export value and the number of transactions shipped by air and by sea within each month. For each firm-month and mode, we record the good (HS6) and destination country of the shipment accounting for the largest export value, which we take to represent that firm-month’s shipment, and we merge in the average maritime transit time from Colombia to each destination, expressed in months.

Second, we merge this exports panel with the monthly export-payments panel described above, defining the monthly inflow as the sum of advance payments, payments received within a year, and longer-term trade credit. We restrict the sample to 2008–2019 and balance, setting export and payment values to zero in months without activity. We then drop firms that never export or never receive any payment. Third, in every month we identify the firm’s most recent shipment and carry forward its characteristics—export value, air value share, good, destination, and maritime transit time. Our outcome is the payment lag, defined as the number of months between a month in which a payment is received and the month of that most recent shipment. To limit the influence of extreme values, we drop payment lags above the 99th percentile.

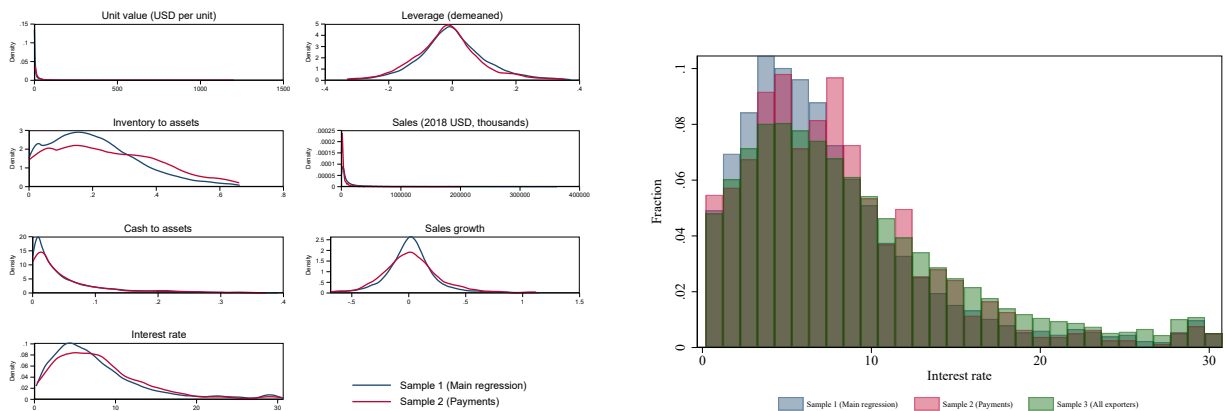
Because inflows are observed at the firm-month level and cannot be matched to individual shipments, we restrict attention to shipments that are isolated in time, so that any subsequent inflow can be unambiguously attributed to a single shipment. For our main estimates (Table 3), we keep each firm’s first shipment together with the inflows received afterward, provided the firm does not ship again for at least twelve months. In the extended sample used in Appendix Table ??, we instead keep any shipment separated from both the previous and the next shipment by at least a year. In both cases the estimation window covers 2009–2018, the regressor *Air* is the value share of the linked shipment carried by air (equal to zero or one for the single-mode shipments that make up the vast majority of the sample), observations are weighted by the export value of the linked shipment, and standard errors are clustered by month.

A.6 Comparing Payments and Exports Transactions data

Our empirical analysis relies on two estimation samples: the main regression sample in Table 4 (Sample 1) and the payment-gap sample in Table 3 (Sample 2). In this section we compare them across the full set of controls used in the paper and, for the interest rate, against the broader population of Colombian exporters (Sample 3), to assess how representative our samples are and whether they differ substantially throughout the exercises on the empirical

section.

Figure A.2: The estimation samples are similar to each other and broadly representative of Colombian exporters



(a) Firm- and shipment-level controls: main vs. payment samples (b) Interest-rate distribution across the three samples

Note: Panel (a) plots kernel density estimates of the firm- and shipment-level controls used in the analysis for the main regression sample (Sample 1) and the payment-gap sample (Sample 2). All variables are measured at the firm-year level, except unit value, which is measured at the export-transaction level. Panel (b) shows the distribution of the annual interest rate for the two estimation samples and for the full sample of Colombian exporters (Sample 3); the histogram uses common bins and is trimmed to the 1st–99th percentiles of the pooled interest rate.

Figure A.2 compares the two estimation samples across the controls included in Table 4. The distributions are very similar for most variables, and especially for those most relevant to our mechanism: demeaned leverage, cash to assets, and the interest rate, with the average interest rate differing by only 0.2 percentage points (7.9% versus 8.1%, $p = 0.38$). The main differences arise in firm size and inventories. The payment sample is composed of relatively smaller firms that hold slightly more inventories relative to assets and ship higher unit-value goods.

Figure A.2b zooms on the interest-rate distribution and adds the full population of Colombian exporters as reference. The three distributions share the same shape and mode, peaking at rates of roughly 4–8%. Importantly, this figure shows that the interest rate on the main and payment samples behaves very similarly and that our estimation samples track the broader exporter population closely. However, the exporters sample displays a slightly heavier right tail: about 21% of exporters face interest rates above 18%, compared with 3.2% and 3.8% in the main and payment samples, respectively. This indicates that our samples are tilted modestly toward firms with somewhat cheaper credit, but the substantial overlap across the bulk of the distribution reinforces that the credit conditions we exploit are representative of those faced by Colombian exporters more generally.

Appendix B Facts appendix

B.1 Additional results on payment gaps by mode of shipping

B.1.1 No difference in advance payment by mode of shipment

We do not find evidence that firms using different transport modes obtain systematically distinct ex-ante working-capital financing from their buyers through advance payments, even after controlling for export size and a rich set of fixed effects. We estimate the difference in the probability of receiving advance payments between air and sea shipments, controlling for export size and a rich set of fixed effects, using the following regression:

$$\text{Adv}_{it} = \alpha_t + \beta_1 \text{Air}_{it} + \beta_2 \ln \text{value}_{it} + \delta_{\text{gap}(i)} + \gamma_{g(i)} + \eta_{d(i)} + \varepsilon_{it}, \quad (\text{B.1})$$

where i is a shipment in month t . Adv_{it} is a dummy for receiving payments recorded as advanced before shipment i in month t . Air_i is a dummy for whether the export was made by air.³⁵ In some specifications, we add fixed effects for the gap between the matched export transaction and the previous or next export transaction, $\delta_{\text{gap}(i)}$. We also control for a good (HS-6) fixed effect ($\gamma_{g(i)}$), destination fixed effect ($\eta_{d(i)}$) and month fixed effect (α_t). We weight each observation by value. Table B.1 shows that the probability of receiving advance payments does not differ statistically between air and sea shipments.³⁶

Table B.1: Air and sea shipments are equally likely to receive advanced payments

	Prob. of receiving advanced payments		
	(1)	(2)	(3)
Air_i	-0.019 (0.027)	-0.005 (0.030)	0.050 (0.035)
$\ln \text{value}_i$	0.007 (0.011)	0.018 (0.013)	-0.020 (0.015)
Observations	8,880	8,875	8,513
t FE	✓	✓	✓
Time to next shipment FE		✓	✓
HS6 and destination country FE			✓

Note: This table displays the results from estimating equation (B.1). All regressions are weighted by export value. Standard errors are clustered at the month level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.1.2 Payment delay regression robustness

Extended sample We expand the results from Table 3 in two ways. First, we include shipments where the share of air is not 0 or 1. Table B.2 presents the results. Second, we include in the estimation sample any shipment happening at least a year after the previous

³⁵In the rare cases where a firm shipped by both air and sea in the same month, we compute the share of air weighted by value.

³⁶Recall that advanced payments are directly identified in the payments dataset, so measuring them does not require matching individual payments to specific shipments

shipment and matching it with the nearest next payment. Table B.3 presents the results. Both tables similarly shows a negative and significant coefficient, consistent with the results found in the main part of the paper.

Table B.2: Payment delay by shipment mode: including mixed-mode shipments

	Time since last export shipment (delay_i)			
	(1)	(2)	(3)	(4)
Air_i	-3.550*** (0.595)	-2.991*** (0.575)	-1.863** (0.902)	0.236 (1.265)
$\ln \text{value}_i$	-1.620*** (0.121)	-1.380*** (0.121)	-0.715*** (0.252)	-0.579** (0.257)
SeaTime_i				3.166 (2.327)
$\text{SeaTime}_i \times \text{Air}_i$				-3.602* (1.981)
Observations	8880	8875	8511	8387
t FE	✓	✓	✓	✓
Time to next shipment FE		✓	✓	✓
HS6 FE			✓	✓
Destination country FE			✓	
Gravity controls				✓

Note: This table displays the results from estimating equation (2) on the sample of first shipments (occurring at least a year before the next shipment), including firms that ship in mixed modes. “Time to next shipment FE” is a fixed effect for the number of months between the export shipment and the next export shipment. Gravity controls include latitude and longitude of the destination country, a dummy for landlocked, contiguousness, past colonial experience and a continent fixed effect. All regressions are weighed by export value. Robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

No value weights We also repeat the results from Table 3 without weighting observations by the value of the transaction. Table B.4 presents the results.

B.1.3 Event-study results

We also estimate an event-study design to characterize the dynamics of export payments following a firm’s initial entry into foreign markets. Our identification strategy exploits variation in the timing of firms’ first export, allowing us to trace how payments evolve in the months before and after this event.³⁷ Formally, we estimate the following specification:

$$Y_{jmt} = \beta_0 + \sum_{s=-3}^{12} \beta_{ms} D_{jm,t-s} + X'_{jtm} \beta + \alpha_j + \alpha_t + \epsilon_{jmt}, \quad (\text{B.2})$$

³⁷As a robustness check, Appendix B.2 reports results using the second export event as the reference point. Each specification entails a trade-off. Using the first export provides cleaner identification of payment dynamics but offers limited scope for additional controls, as many firms export only once. Using the second export increases variation and allows for richer controls, such as destination and product, though at the cost of weaker identification of the initial timing of payments.

Table B.3: Payment delay by shipment mode: isolated shipments

	Time since last export shipment (delay_i)			
	(1)	(2)	(3)	(4)
Air_i	-1.781*** (0.521)	-1.932*** (0.437)	-1.278** (0.629)	0.698 (0.918)
$\ln \text{value}_i$	-1.127*** (0.115)	-1.150*** (0.100)	-0.779*** (0.193)	-0.656*** (0.199)
SeaTime_i				1.536 (1.928)
$\text{SeaTime}_i \times \text{Air}_i$				-3.298** (1.546)
Observations	14379	14368	14001	13790
t FE	✓	✓	✓	✓
Time to next shipment FE		✓	✓	✓
HS6 FE			✓	✓
Destination country FE			✓	
Gravity controls				✓

Note: This table displays the results from estimating equation (2) on the sample of isolated shipments, i.e. shipments occurring at least a year after the previous shipment and at least a year before the next one, restricted to firms shipping in a single mode (all air or all sea). “Time to next shipment FE” is a fixed effect for the number of months between the export shipment and the next export shipment. Gravity controls include latitude and longitude of the destination country, a dummy for landlocked, contiguousness, past colonial experience and a continent fixed effect. All regressions are weighed by export value. Robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

where Y_{jmt} denotes the dollar value of export payments received by firm j , shipping by mode m , in month t . The variable $D_{jm,t-s}$ is an event-time indicator equal to one if month t is s months relative to the first export (with $s = 0$ denoting the first shipment). Motivated by the data, which show that payments often occur up to a quarter before the recorded first export, we allow for leads of up to three months. We also track payments for twelve months following the event.³⁸ We include as controls, X_{jtm} the value of subsequent exports (to isolate dynamics specific to the initial export). We also include firm (α_j) and time (α_t) fixed effects.³⁹

To isolate systematic differences in payment delays across transport modes, we restrict the sample to firms that export exclusively by sea or exclusively by air and we estimate two separate regressions, one for each shipping mode, and recover distinct sets of event-time coefficients β_{ms} , and show them in figure B.1. Panel (a) reports results for air shipments, while Panel (b) reports results for sea shipments. The vertical axis plots the estimated coefficients, and the horizontal axis denotes event time s , measured in months relative to the first export. We omit the month immediately preceding the first export, so all coefficients are interpreted relative to that baseline period.

Our results show that firms shipping by air receive their payments between one month

³⁸Table B.5 shows summary statistics for the variables relevant to this exercise.

³⁹When using the second time a firm exports, we can control by the main product exported in a given month, and the city of destination.

Table B.4: Payment delay by shipment mode: unweighted estimates

	Time since last export shipment (delay_i)			
	(1)	(2)	(3)	(4)
Air_i	-0.888*** (0.286)	-1.045*** (0.283)	-1.029* (0.556)	0.254 (0.738)
$\ln \text{value}_i$	-0.867*** (0.0952)	-0.805*** (0.0925)	-0.910*** (0.155)	-0.840*** (0.152)
SeaTime_i				2.224* (1.229)
$\text{SeaTime}_i \times \text{Air}_i$				-2.362** (1.190)
Observations	8862	8857	8491	8369
t FE	✓	✓	✓	✓
Time to next shipment FE		✓	✓	✓
HS6 FE			✓	✓
Destination country FE			✓	
Gravity controls				✓

Note: This table displays the results from estimating equation (2) on the sample of first shipments (occurring at least a year before the next shipment) restricted to firms shipping in a single mode (all air or all sea), estimated without weights. “Time to next shipment FE” is a fixed effect for the number of months between the export shipment and the next export shipment. Gravity controls include latitude and longitude of the destination country, a dummy for landlocked, contiguousness, past colonial experience and a continent fixed effect. Regressions are unweighted. Robust standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

before and up one month after they export, while firms shipping by sea receive payments during the first two quarters after the export, with an up-think during months five and six. It is relevant to mention that all other coefficients are not statistically significant, reinforcing the idea that firms do not receive disproportionately more anticipated payments. This results suggests that access to credit might be fundamental in choosing the mode of transportation given that there is a mismatch between the shipment and the payments. Firms with more access to credit could afford a cheaper yet slower mode of transportation, while firms without access to credit will need to choose a more expedite more – air– in order to finance their obligations.

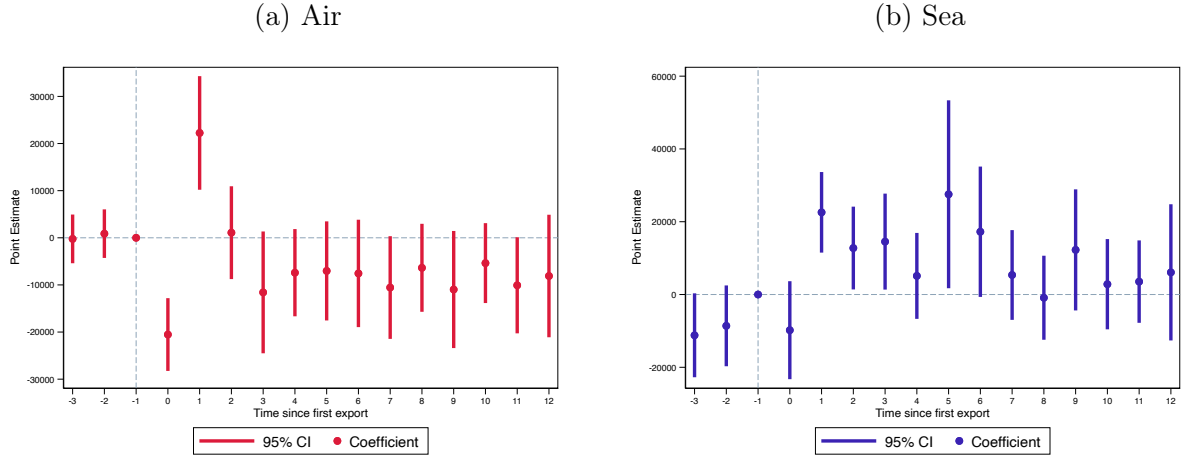
B.1.4 Implied interest rate substituting financing by air

This appendix uses the payment delay estimates from Section 3 to compute the implied interest rate at which a firm is indifferent between shipping by sea and waiting for payment, or paying the freight premium to ship by air and collecting revenue sooner. The indifference condition equates the total cost of each option:

$$(V + c_a)(1 + R)^{\bar{\tau}_a/12} = (V + c_s)(1 + R)^{\bar{\tau}_s/12}, \quad (\text{B.3})$$

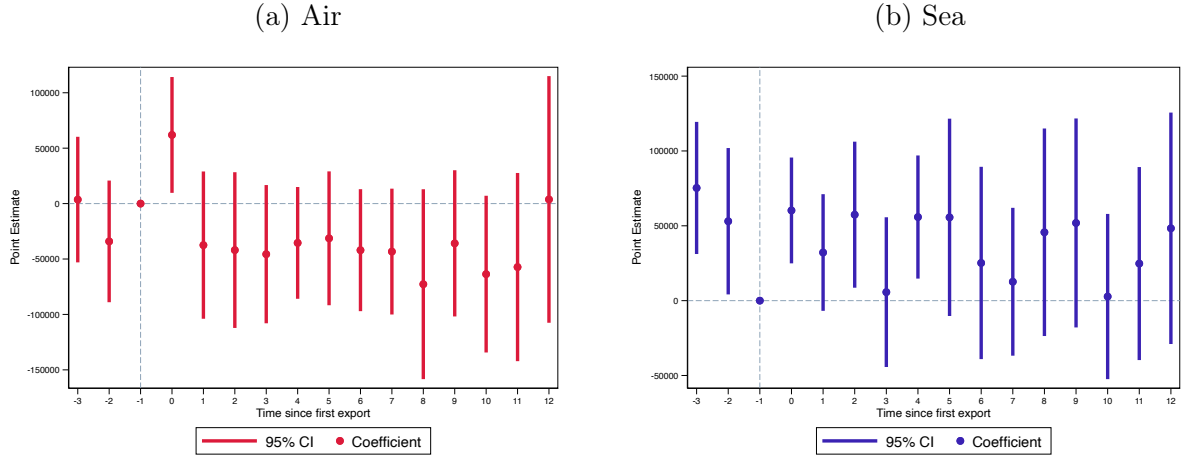
where V is export value, c_a and c_s are freight costs under air and sea, and $\bar{\tau}_a$, $\bar{\tau}_s$ are the respective payment delays in months. Since the production lag cancels, this yields an exact

Figure B.1: Payment delays are longer for sea shipments than for air shipments



Note: This figure plots the estimated coefficients β_{ms} from equation B.2. The vertical axis measures the dollar change in export payments relative to the month prior to the first shipment, while the horizontal axis reports event time in months before and after the initial export. Panel (a) (left) displays results for air shipments (red dots), and Panel (b) (right) displays results for sea shipments (blue dots). Vertical bars denote 95% confidence intervals based on robust standard errors clustered at the firm-month level. The air-shipment regression is estimated with 222,541 observations, and the sea-shipment regression with 166,711 observations.

Figure B.2: Payment delays are longer for sea shipments than for air shipments using the second export



Note: This figure plots the estimated coefficients β_{ms} from equation B.2 and adding mode good and city of destination fixed effects. The vertical axis measures the dollar change in export payments relative to the month prior to the second shipment, while the horizontal axis reports event time in months before and after the initial export. Panel (a) (left) displays results for air shipments (red dots), and Panel (b) (right) displays results for sea shipments (blue dots). Vertical bars denote 95% confidence intervals based on robust standard errors clustered at the firm-month level. The air-shipment regression is estimated with 222,541 observations, and the sea-shipment regression with 166,711 observations.

Table B.5: Summary Statistics - Event Studies

	Mean	Std. Dev	Pctile. 5	Median	Pctile. 95	N
Total payments ^(k)	31	480	0	0	53	135501
Short term payments ^(k)	19	300	0	0	35	135501
Advanced payments ^(k)	12	296	0	0	12	135501
Long term payments ^(k)	0.04	2	0	0	0	135501

Note: This table reports summary statistics at the firm, export transaction, and monthly export-relationship levels. Superscript ^(k) denotes variables measured in 2018 USD thousands. Pctile. 5, Median, and Pctile. 95 denote the 5th percentile, the median, and the 95th percentile of each variable, respectively. The instrument corresponds to the weighted average of a firm’s exposure to banks with stronger Tier 1 capital ratios. Source: *SuperSociedades*, *DIAN*, *Formato 341*, *Banco de la República* and author’s elaboration.

expression for the implied annual rate:

$$R = \left(\frac{V + c_a}{V + c_s} \right)^{12/\bar{\tau}_s} - 1, \quad (\text{B.4})$$

where the ratio $(V + c_a)/(V + c_s) = 1 + \hat{\beta}_{\text{Air}}$ is estimated in Section 3

Applying the payment delay bounds from Tables B.6, 3 and annualizing yields an implied rate range of 17% to 32%.⁴⁰ Figure ?? shows that this range is well within the support of the interest rate distribution observed in our sample, confirming that the substitution mechanism is relevant for a meaningful share of Colombian exporters. Notably, when we think of the whole export sample, we find that 43% of the firms have rates above 18%, making this mechanism binding. However, as we discuss in Section 3, interest rates are one margin along which credit conditions vary; lenders also adjust loan quantities, maturities, and collateral requirements, and our main results capture this broader variation.

⁴⁰The longer delay produces the lower implied rate: with more time before payment, the per-period cost of waiting is smaller, so a lower interest rate suffices to rationalize the air freight premium.

Table B.6: Interest rate that substitutes financing by air within the same good

	Transportation Cost to export value			
	(1)	(2)	(3)	(4)
	Payments Sample	Full sample		
Air	0.051*	0.061**	0.053*	0.056*
	(0.031)	(0.029)	(0.030)	(0.031)
ImpliedRate, %	36.12-18.45	43.92-22.13	37.70-19.20	40.01-20.29
Observations	101859	7126015	7123175	7321494
odmy_hs8 FE	✓	✓	✓	
odmy_hs6 FE				✓
Firm FE			✓	✓

Note: This table reports the implied interest rate at which firms are indifferent between shipping by air and financing sea shipments. The first row reports the estimated coefficient from regressing transportation costs divided by export value on an air dummy. The second row reports the implied annual interest rate, calculated using the exact expression $R = (1 + \hat{\beta}_{\text{Air}})^{12/\bar{\tau}_s} - 1$ and the payment delay estimates from columns (1) and (3) of Table 3 (1.949 and 3.55 months, respectively). Column (1) uses the payment-matched sample; columns (2)–(4) use the full sample of exports. odmy_hs8 denotes origin region, destination city, month, year, and HS8 product classification; odmy_hs6 denotes the same at the HS6 product level. Standard errors clustered at the firm level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.2 Firms with higher access to finance use less air shipments

B.2.1 Robustness checks

In this section, we present several exercises that confirm the robustness of our empirical findings.

First, we examine whether access to finance on the extensive margin fully explains the shifts in the probability of using air, rather than the intensive margin measures explored in Section 3. Finding a prevalence of the extensive margin in changing the probability of shipping goods by air would imply that entry into the credit market plays a higher role than the intensity of credit usage on transportation choices. Formally, we estimate the following specification:

$$\begin{aligned} \text{Air}_{odgjt} = & \beta_0 + \beta_1 \text{BankAccess}_{jt} + \sum_k \beta_k \text{CreditMeasure}_{jt} \cdot \text{BankAccess}_{jt} \\ & + \beta_3 \log(\text{unitValue})_{odgjt} + \alpha_{ts} + \alpha_{odgt} + X'_{jt}\beta + \epsilon_{odgjt} \end{aligned} \quad (\text{B.5})$$

where BankAccess_{jt} is a dummy equal to one if firm j has at least one active loan on year t and $\text{CreditMeasure}_{jt}$ includes the intensive margin measures analyzed in Section 3. The rest of the equation's elements are homologous to the ones in Equation (3). Table B.7 reports the results and reveals that bank access on the extensive margin does not significantly alter the probability of using air transport. The interaction terms between bank access and the intensive margin measures retain the same signs as in our main estimates, and the interaction with interest payments to sales remains statistically significant, consistent with the conclusion that greater access to finance reduces the likelihood that firms use air mode.

Table B.7: Higher access to banking credit reduces the probability of shipping by air

	Prob. of shipping by air			
	(1)	(2)	(3)	(4)
Bank access dummy	-0.016 (0.021)	-0.009 (0.022)	-0.017 (0.022)	-0.008 (0.020)
Bank access * Int. payments to sales	-0.026** (0.013)			
Bank access * Int. rate		0.007 (0.004)		
Bank access * Total debt			-0.011 (0.007)	
Bank access * More loans				-0.011 (0.010)
Log(Unit value)	0.021*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.021*** (0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
R^2	0.760	0.760	0.760	0.760
N	4,568,834	4,568,834	4,568,834	4,568,834

Note: This table shows the extensive margin results for the probability of shipping by air. Bank access dummy is equal to one if firm j has at least one active loan in year t . Columns present alternative measures for credit access on the intensive margin: (1) Bank access dummy and interaction with interest payments to sales; (2) Bank access dummy and interaction with the interest rate; (3) Bank access dummy and interaction with total debt; and (4) Bank access dummy with variables for active loans above the median. Unit value measures the shipment value divided by the number of units in logs. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth and FDI receipt in the extensive, which are lagged by one year. Robust standard errors clustered at the firm level. *** : $pval < 0.01$, ** : $pval < 0.05$, * : $pval < 0.1$.

Second, we test the robustness of our estimates on the relationship between credit access on the intensive margin and transportation mode by varying the fixed effects included in the specifications. Specifically, we consider multiple combinations of fixed effects including origin, destination, good and month separately, interacted origin-destination-good-month, year-sector, firm of origin, and even firm of destination effects, as documented in Tables B.8 and B.9 by checkmarks. The estimated equation is the same as Equation (3) from the main text, with the key difference being the set of fixed effects used across specifications. Controls remain identical to those in the main analysis. Across all these alternative setups, Tables B.8 and B.9 show that our key IV estimates remain stable in sign and magnitude, and statistically significant in most specifications, confirming the robustness of our main findings.

Table B.8: IV results on interest payments and interest rates are robust to different specifications

	(1)	(2)	(3)	(4)	Prob. of shipping by air		(7)	(8)	(9)	(10)
					(5)	(6)				
Int. payments to sales	0.204 (0.338)	-0.302** (0.121)	-0.351*** (0.134)	-0.198* (0.114)	-0.143 (0.091)					
Int. rate						-0.023 (0.095)	0.087*** (0.032)	0.091*** (0.035)	0.033 (0.022)	0.031 (0.021)
Log(Unit value)	0.010 (0.007)	0.035*** (0.004)	0.019*** (0.003)	0.014*** (0.005)	0.012*** (0.004)	0.011 (0.007)	0.034*** (0.004)	0.019*** (0.003)	0.013*** (0.005)	0.012*** (0.005)
City Origin FE		✓					✓			
City Destination FE		✓					✓			
Good FE		✓					✓			
Month FE		✓					✓			
Origin-Dest-Good-Month			✓	✓	✓			✓	✓	✓
Year-Sector FE				✓	✓				✓	✓
Firm Origin FE				✓					✓	
Firm Destination FE					✓					✓
Firm level Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	4,863,186	4,861,440	4,595,841	4,558,741	4,518,554	4,602,272	4,600,578	4,346,753	4,310,053	4,272,009

Note: This table shows the robustness of IV results on interest payments to sales and interest rate to different specifications. Columns (1) and (6) have no fixed effects; columns (2) and (7) include origin, destination, good and month fixed effects individually; columns (3) and (8) have origin-destination-good-month fixed effects; columns (4) and (9) add firm of origin, in addition to origin-destination-good-month fixed effects; and columns (5) and (10) have origin-destination-good-month and firm of destination fixed effects. Unit value measures the shipment value divided by the number of units in logs. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth and FDI receipt in the extensive, which are lagged by one year. Robust standard errors clustered at the firm level. *** : $pval < 0.01$, ** : $pval < 0.05$, * : $pval < 0.1$.

Third, we re-estimate the same specification as in Table 4 defining goods at the more disaggregated HS8 level. Table B.10 shows that the estimated coefficients are very close in sign and magnitude to the baseline, and our key results remain statistically significant, indicating that our findings are not driven by the level of product aggregation.

Fourth, we re-estimate our main intensive-margin specifications clustering standard errors at the firm-year rather than the firm level, which relaxes the assumption that a firm's shipments are correlated across years. Table B.11 shows that the OLS estimates become overall more significant under this less conservative clustering, while our key IV coefficients retain their sign and statistical significance, confirming that, once we account for endogeneity, our inference is not sensitive to the choice of clustering.

Fifth, we split the intensive margin measures into short- and long-term components to assess whether the effects of financial access differ over time. This distinction is important because it allows us to identify the time horizon that primarily affects the probability of shipping by air for each credit access measure. Results are displayed in Table B.12. We

Table B.9: IV results on total debt and number of active loans are robust to different specifications

	(1)	(2)	(3)	(4)	Prob. of shipping by air		(7)	(8)	(9)	(10)
					(5)	(6)				
Total debt	0.025 (0.050)	-0.044*** (0.015)	-0.047*** (0.015)	-0.017* (0.009)	-0.015 (0.009)					
More loans						0.152 (0.311)	-0.264** (0.113)	-0.356** (0.170)	-0.139 (0.087)	-0.108 (0.078)
Log(Unit value)	0.011 (0.007)	0.036*** (0.004)	0.020*** (0.003)	0.014*** (0.005)	0.012*** (0.004)	0.010 (0.008)	0.035*** (0.004)	0.018*** (0.004)	0.013*** (0.004)	0.011*** (0.004)
City Origin FE		✓					✓			
City Destination FE		✓					✓			
Good FE		✓					✓			
Month FE		✓					✓			
Origin-Dest-Good-Month			✓	✓	✓			✓	✓	✓
Year-Sector FE				✓	✓				✓	✓
Firm Origin FE				✓					✓	
Firm Destination FE					✓					✓
Firm level Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	4,870,974	4,869,226	4,603,270	4,566,166	4,525,938	4,873,346	4,871,595	4,605,489	4,568,339	4,528,090

Note: This table shows the robustness of IV results on total debt and number of loans above the median to different specifications. Columns (1) and (6) have no fixed effects; columns (2) and (7) include origin, destination, good and month fixed effects individually; columns (3) and (8) have origin-destination-good-month fixed effects; columns (4) and (9) add firm of origin, in addition to origin-destination-good-month fixed effects; and columns (5) and (10) have origin-destination-good-month and firm of destination fixed effects. Unit value measures the shipment value divided by the number of units in logs. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth and FDI receipt in the extensive, which are lagged by one year. Robust standard errors clustered at the firm level. *** : $pval < 0.01$, ** : $pval < 0.05$, * : $pval < 0.1$.

observe consistent signs for the intensive margin coefficients both in the short- and long-term, but the effects are generally weaker or insignificant in the short term, reflecting how credit cost and access differ in their impact on transport mode choice at different horizons.

Finally, we estimate how the effect of sea shipping time on the probability of using air depends on firms' access to bank credit, using IV to purge the credit variables of endogeneity. With this specification, our aim is to test whether credit conditions still shape transport choices even after directly controlling for a key transaction characteristic—how long it takes to ship by sea—and thus complements the main analysis by ruling out that sea time alone explains the shift from sea to air. Table B.13 shows that the coefficients of the interaction-term are statistically significant and their magnitudes and signs align with the findings in the main text. This pattern implies that for firms with better or cheaper access to credit (higher interest payments to sales, more debt, more loans, and lower interest rates), the marginal effect of longer sea shipping time on switching to air is attenuated, whereas firms facing higher interest rates are more sensitive to delays and more likely to substitute into air when sea is slow. In other words, even conditional on sea shipping time and rich fixed effects, intensive-margin credit conditions remain economically and statistically relevant for transport-mode choice.

B.2.2 IV assumptions

To isolate exogenous variation in credit supply, we employ an instrumental variable (IV) strategy based on the Tier 1 capital ratio of banks to which each firm is exposed. The Tier 1 capital ratio—defined as a bank's core equity capital divided by its total risk-weighted assets—captures bank-level capitalization and risk exposure, and is a standard measure of financial robustness used to proxy for credit supply conditions (see, e.g., Jiménez et al., 2017; Gropp et al., 2019; Correia et al., 2024; Ormazabal et al., 2025).

The instrument for firm j at time t is constructed as the weighted average Tier 1 capital

Table B.10: Main results - Fixed effects with 8-digit HS code

	Prob. of shipping by air			
	(1)	(2)	(3)	(4)
Panel A: Baseline Estimates				
Int. payments to sales	-0.027** (0.013)			
Int. Rate		0.007 (0.005)		
Total debt			-0.013* (0.007)	
More loans				-0.013 (0.011)
Log(Unit value)	0.020*** (0.004)	0.020*** (0.004)	0.021*** (0.004)	0.020*** (0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
N	4,515,905	4,269,367	4,523,301	4,525,467
Panel B: IV Estimates				
Int. payments to sales	-0.296** (0.115)			
Int. Rate		0.077** (0.031)		
Total debt			-0.038*** (0.013)	
More loans				-0.275** (0.134)
Log(Unit value)	0.020*** (0.004)	0.019*** (0.004)	0.021*** (0.004)	0.019*** (0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
N	4,515,905	4,269,367	4,523,301	4,525,467

Note: This table displays the estimated coefficients for the impact of banking credit access on the likelihood of shipping by air, with goods defined at the HS8 level. Panel A shows baseline estimates for equation 3 using alternative measures for β_1 : (1) Interest payments to sales (standardized); (2) Interest rate (standardized); (3) Total debt (standardized); and (4) Dummy for having a number of loans above the median. Panel B reports IV estimates, where the instrument is defined by the firm's exposure to healthier banks (see equation 4). The variable "Unit value" measures the shipment value divided by the number of units in logs. In both panels, total debt is total outstanding bank loans, interest payments to sales is annual interest payments to banks over first sales, and "more loans" is a dummy for firms above the sample median number of bank loans. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth and FDI receipt in the extensive, which are lagged by one year. Robust standard errors are clustered at the firm level. *** : $p < 0.01$, ** : $p < 0.05$, * : $p < 0.1$.

ratio of the banks in firm j 's credit portfolio:

$$\text{Tier1Exposure}_{jt} = \sum_b w_{jb,t-1} \text{Tier1Ratio}_{b,t-1}, \quad (\text{B.6})$$

where $w_{jb,t-1}$ denotes the share of credit that firm j obtained from bank b in period $t - 1$. This weighting structure captures the intensity of firm-bank relationships while mitigating simultaneity between firm-level outcomes and contemporaneous bank conditions.

Table B.11: Main results - Standard errors at the firm-year level

	Prob. of shipping by air			
	(1)	(2)	(3)	(4)
Panel A: Baseline Estimates				
Int. payments to sales	-0.026*** (0.009)			
Int. Rate		0.007* (0.004)		
Total debt			-0.011*** (0.003)	
More loans				-0.011** (0.005)
Log(Unit value)	0.021*** (0.002)	0.020*** (0.002)	0.021*** (0.002)	0.021*** (0.002)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
N	4,559,233	4,310,538	4,566,662	4,568,834
Panel B: IV Estimates				
Int. payments to sales	-0.288*** (0.077)			
Int. Rate		0.075*** (0.022)		
Total debt			-0.037*** (0.009)	
More loans				-0.257*** (0.093)
Log(Unit value)	0.020*** (0.002)	0.020*** (0.002)	0.021*** (0.002)	0.019*** (0.002)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
N	4,559,233	4,310,538	4,566,662	4,568,834

Note: This table displays the estimated coefficients for the impact of banking credit access on the likelihood of shipping by air. Panel A shows baseline estimates for equation 3 using alternative measures for β_1 : (1) Interest payments to sales (standardized); (2) Interest rate (standardized); (3) Total debt (standardized); and (4) Dummy for having a number of loans above the median. Panel B reports IV estimates, where the instrument is defined by the firm's exposure to healthier banks (see equation 4). The variable "Unit value" measures the shipment value divided by the number of units in logs. In both panels, total debt is total outstanding bank loans, interest payments to sales is annual interest payments to banks over first sales, and "more loans" is a dummy for firms above the sample median number of bank loans. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth and FDI receipt in the extensive, which are lagged by one year. Robust standard errors are clustered at the firm-year level. *** : $p < 0.01$, ** : $p < 0.05$, * : $p < 0.1$.

The validity of the instrument rests on two key assumptions. First, the instrument's relevance is ensured if differences in bank capitalization translate into differential lending capacity, i.e., $\text{Cov}(\text{Tier1Exposure}_{bt}, \text{CreditSupply}_{bt}) \neq 0$. Ample empirical evidence shows that better-capitalized banks are less constrained by regulatory or funding limits and therefore expand credit more readily during downturns. These findings support a direct supply-side channel through which variation in bank Tier 1 capital affects lending behavior.

Exogeneity requires that conditional on controls, the Tier 1 ratios of banks reflecting past

Table B.12: Main empirical results in the short and long term - IV

	(1)	(2)	(3)	Prob. of shipping by air		(6)	(7)	(8)
				(4)	(5)			
Int. paym. to sales, short term	-0.356*							
	(0.209)							
Int. paym. to sales, long term		-0.370***						
		(0.140)						
Int. rate, short term			0.159					
			(0.140)					
Int. rate, long term				0.100**				
				(0.046)				
Total debt, short term					-0.107**			
					(0.050)			
Total debt, long term						-0.042***		
						(0.015)		
More loans, short term							-1.544	
							(3.062)	
More loans, long term								-0.211**
								(0.104)
Log(Unit value)	0.019***	0.020***	0.018***	0.019***	0.020***	0.021***	0.017*	0.020***
	(0.005)	(0.004)	(0.005)	(0.004)	(0.005)	(0.004)	(0.009)	(0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓	✓	✓	✓	✓
<i>N</i>	3,507,485	4,432,905	3,504,772	3,869,468	3,513,645	4,434,707	4,568,834	4,568,834

Note: This table reports the IV estimated coefficients from regressions that include alternative measures of financial variables. Each specification includes Origin-Destination-Good-Month and Year-Sector fixed effects, as well as firm-level controls. Robust standard errors are clustered at the firm level. *** : $p < 0.01$, ** : $p < 0.05$, * : $p < 0.1$.

Table B.13: Sea Shipping Time, Credit Access, and the Probability of Shipping by Air - IV

	Prob. of shipping by air			
	(1)	(2)	(3)	(4)
Sea shipping time	0.002	0.012***	0.013***	0.034**
	(0.004)	(0.002)	(0.003)	(0.014)
Sea time × Int. payments to sales	-0.051**			
	(0.021)			
Sea time × Int. rate		0.017**		
		(0.008)		
Sea time × Total debt			-0.005***	
			(0.002)	
Sea time × More loans				-0.041*
				(0.021)
Log(Unit value)	0.020***	0.019***	0.021***	0.019***
	(0.004)	(0.004)	(0.004)	(0.004)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
<i>N</i>	4,514,433	4,267,522	4,521,862	4,524,022

Note: This table reports the IV estimates of the impact of sea shipping time and its interaction with measures of banking credit access on the likelihood of shipping by air. Interaction terms combine sea shipping time with alternative measures of credit access: interest payments to sales (standardized), interest rate (standardized), total debt (standardized), and a dummy for having a number of loans above the median. Unit value is the log shipment value divided by the number of units. Firm-level controls include sales, inventory-to-assets, leverage, cash to assets, sales growth, and FDI receipt in the extensive margin, all lagged one year. Robust standard errors are clustered at the firm level. *** : $p < 0.01$, ** : $p < 0.05$, * : $p < 0.1$.

profitability, capital management, and regulatory adjustments are orthogonal to firm-level shocks to credit demand or performance. Under these conditions, variation in the constructed

exposure measure identifies shifts in firm-level credit supply driven by bank balance sheet strength rather than by firm-specific credit demand. The literature argues that this condition is plausible when capital variation is driven by bank-level fundamentals and regulatory forces rather than by contemporaneous borrower outcomes, especially when capital measures are lagged and firm characteristics are controlled for (e.g., Jiménez et al., 2017; Correia et al., 2024). This assumption is particularly compelling in the Colombian context, where banks are large, diversified, and largely non-specialized, with lending portfolios spanning a wide range of firms and sectors. As a result, individual firms account for a negligible share of banks’ balance sheets, limiting the scope for firm-specific shocks to influence bank capitalization (Aristizábal-Ramírez and Posso, 2021). Under these conditions, variation in the constructed exposure measure identifies shifts in firm-level credit supply driven by bank balance sheet strength rather than by firm-specific credit demand.

Finally, we demonstrate the credibility and robustness of our instrumental variables strategy by presenting the first stage, reduced form, and instrument balancing results. In the first stage of the IV approach, we assess the strength and relevance of our instrument—firms’ differential exposure to the health of their lending banks—by estimating its effect on the endogenous credit access variables used in the main analysis. Intuitively, firms more exposed to healthier banks are expected to have improved credit access, as healthier banks are more able to lend and provide favorable credit terms. Therefore, we anticipate—and indeed find—positive first-stage coefficients for variables such as interest payments to sales, total debt and loans above the median, and negative coefficients for interest rates. Importantly, the F statistics reported in Table B.14 are above conventional thresholds, especially for interest payments to sales, the interest rate, and total debt, which indicates a strong first stage.

Table B.14: First Stage Estimates

	(1) Int. payments to sales	(2) Int. Rate	(3) Total debt	(4) More loans
Z	0.025*** (0.006)	-0.115*** (0.025)	0.198*** (0.028)	0.028*** (0.011)
Origin-Dest-Good-Month FE	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓
F statistic	16.03	21.03	50.67	7.160
N	4,559,233	4,310,538	4,566,662	4,568,834

Note: This table presents the first stage regression coefficients. Where Z is the firm’s exposure to healthier banks as defined in equation 4). Total debt represents total outstanding bank loans; interest payments-to-sales measures annual interest payments to banks over sales; rate is the average yearly bank interest rate; “more loans” is a dummy equal to one for firms above the sample median number of bank loans. Robust standard errors are clustered at the firm level. *** $p < 0.01$; ** denotes $p < 0.05$; * denotes $p < 0.1$.

The reduced form and instrument balancing provide additional validation for our IV strategy by examining the direct relationship between the instrument and the outcome, as well as assessing the exogeneity of the instrument. Table B.15 shows statistically meaningful reduced form effects across most sets of fixed effects, including our preferred specification. Importantly, these relationships align with the theoretical and empirical expectations, since

we find negative and significant associations between firms' exposure to healthier banks and the probability of shipping by air. Meanwhile, Table B.16 indicates that the instrument is well balanced across key covariates, which supports the assumption of independence from the error term.

Table B.15: IV Reduced Form

	Prob. of shipping by air					
	(1)	(2)	(3)	(4)	(5)	(6)
Z	0.005 (0.011)	-0.009*** (0.003)	-0.009*** (0.003)	-0.007*** (0.003)	-0.004* (0.002)	-0.003 (0.002)
Log(Unit value)	0.012* (0.007)	0.036*** (0.004)	0.020*** (0.003)	0.021*** (0.004)	0.014*** (0.005)	0.012*** (0.004)
City Origin FE		✓				
City Destination FE		✓				
Good FE		✓				
Month FE		✓				
Origin-Dest-Good-Month			✓	✓	✓	✓
Year-Sector FE				✓	✓	✓
Firm Origin FE					✓	
Firm Destination FE						✓
Firm level Controls	✓	✓	✓	✓	✓	✓
R^2	0.095	0.618	0.759	0.760	0.784	0.820
N	4,873,346	4,871,595	4,605,489	4,568,834	4,568,339	4,528,090

Note: This table presents reduced form regression coefficients, reporting the effect of the instrument (Z), defined as firm exposure to healthier banks (see equation 4), directly on the outcome variables of interest. Total debt is the firm's total bank loans; interest payments-to-sales measures annual interest payments to banks over sales; rate is the average yearly bank interest rate; "more loans" is a dummy equal to one if the firm has above-median number of bank loans. Each column represents a different specification, given by the selection of fixed effects. All regressions include the baseline set of firm controls. Robust standard errors are clustered at the firm level. *** $p < 0.01$; ** denotes $p < 0.05$; * denotes $p < 0.1$.

Table B.16: Compliers Structure

	(1) $X * \text{Unit value}$	(2) $X * \text{Demeaned leverage}$	(3) $X * \text{Invent. to assets}$	(4) $X * \text{Initial sales}$	(5) $X * \text{Cash to assets}$	(6) $X * \text{Sales growth}$	(7) $X * \text{Dummy: Received FDI}$
Z	5.997 (4.880)	-0.003*** (0.001)	0.005*** (0.002)	0.251*** (0.066)	0.001*** (0.000)	0.072 (0.066)	0.003** (0.001)
Unit value		-0.000 (0.000)	-0.001 (0.001)	-0.022 (0.034)	-0.000 (0.000)	-0.041 (0.037)	0.000 (0.000)
Demeaned leverage	71.966** (36.599)		-0.009 (0.013)	0.137 (0.389)	0.006 (0.004)	0.172 (0.448)	-0.032* (0.019)
Invent. to assets	-80.295 (64.984)	-0.006 (0.006)		0.373 (0.458)	0.011*** (0.003)	0.235 (0.430)	-0.014 (0.011)
Initial sales	-2.476 (3.766)	0.003** (0.001)	0.003 (0.003)		0.001** (0.001)	-0.023 (0.025)	0.001 (0.003)
Cash to assets	-146.080* (77.047)	0.025 (0.018)	-0.026 (0.022)	-2.612*** (0.953)		-0.967 (0.781)	0.041 (0.025)
Sales growth	4.307 (3.531)	0.001 (0.004)	0.010 (0.009)	0.420 (0.360)	0.000 (0.000)		-0.002 (0.001)
Dummy: Received FDI	10.584 (13.800)	-0.006** (0.003)	-0.003 (0.004)	-0.063 (0.246)	0.002** (0.001)	-0.203 (0.189)	
Origin-Dest-Good-Month FE	✓	✓	✓	✓	✓	✓	✓
Year_Sector FE	✓	✓	✓	✓	✓	✓	✓
Firm level Controls	✓	✓	✓	✓	✓	✓	✓
R^2	0.577	0.330	0.467	0.471	0.403	0.214	0.431
N	4,559,233	4,559,233	4,559,233	4,559,233	4,559,233	4,559,233	4,559,233

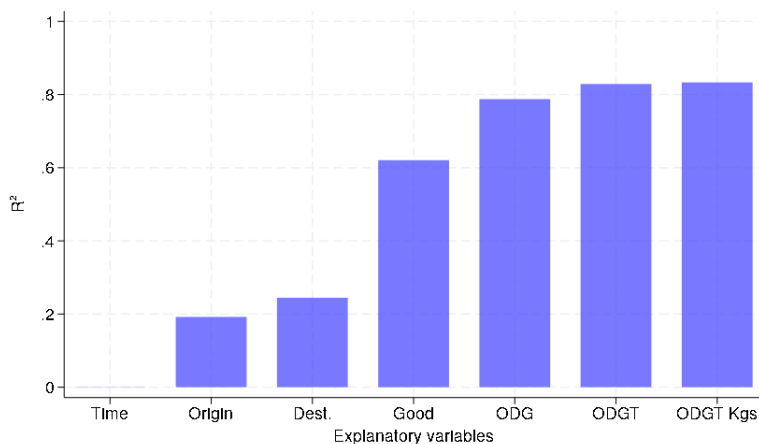
Note: This table reports the distribution of baseline firm characteristics in the effective population identified by our instrumental variable design (i.e., compliers), following Hull (2025). X represents interest payments to sales. All specifications include the same controls and fixed effects as in equation 3, and robust standard errors are clustered at the firm level. *** $p < 0.01$; ** denotes $p < 0.05$; * denotes $p < 0.1$.

B.3 Additional facts on air shipping determinants

B.3.1 “Nature” determinants

Figure B.3 examines how well shipment characteristics typically associated with the “nature” of exports—such as origin, destination, and product—explain firms’ probability of exporting by air. We regress this probability on these shipment-specific fixed effects and track the R-squared. The fact that the R-squared never approaches one indicates that these observable traits alone cannot fully account for the decision to ship by air. This leaves a substantial unexplained margin, suggesting the presence of unobserved factors that influence transport mode choice. We argue that one key driver partially filling this gap is financial constraints faced by firms, which motivates this paper’s analysis.

Figure B.3: Nature cannot explain all air shipments



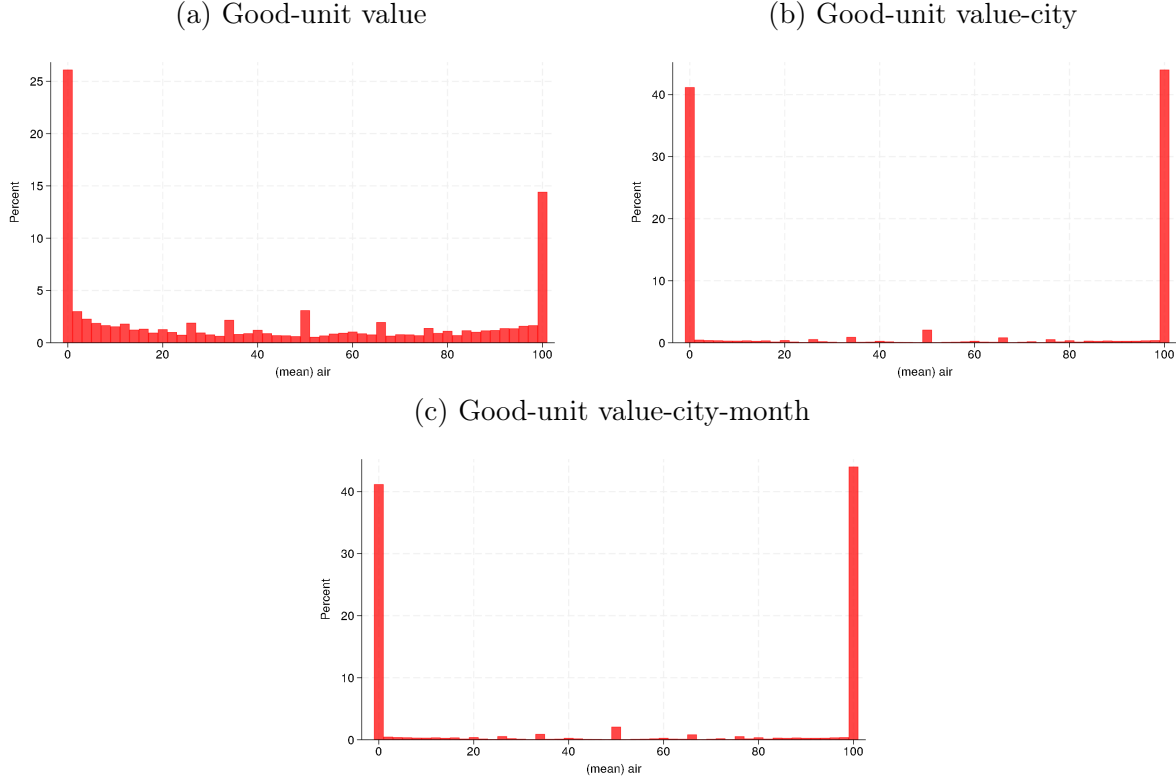
Note: Each bar shows the R^2 of regressing the probability of shipping by air in a set of controls. From left to right the regressions include: Time fixed effects, city of origin fixed effects, city of destination fixed effects, goods at the HS6-digit level fixed effects, City of origin-destination-good fixed effects (ODG), City of origin-destination-good-time fixed effects (ODGT), and City of origin-destination-good-time fixed effects and weight in kilograms by shipment (ODGT Kgs).

B.3.2 Additional facts on air intensity by good

Figure B.4 complements the evidence offered in Fact 3 by summarizing the distribution of air at a more disaggregated level. Panel A shows the distribution for goods with similar unit values, while Panel B focuses on shipments with a comparable value arriving in the same destination city. The results show that, even when controlling for these very specific characteristics, there is still a substantial within-good substitutability: a large share of observations exhibit intermediate air use. This finding reinforces the main text’s argument: even for identical products and routes, firms act upon a range of incentives and constraints that drive heterogeneity in export mode selection.

Additionally, Figure B.5 complements the previous histogram by displaying the cumulative share of air shipments across several dimensions: transactions, goods, value, CO2 emissions, and weight. The graph evidences considerable heterogeneity on air usage across

Figure B.4: Good transport mode mix controlling for unit value and city of destination



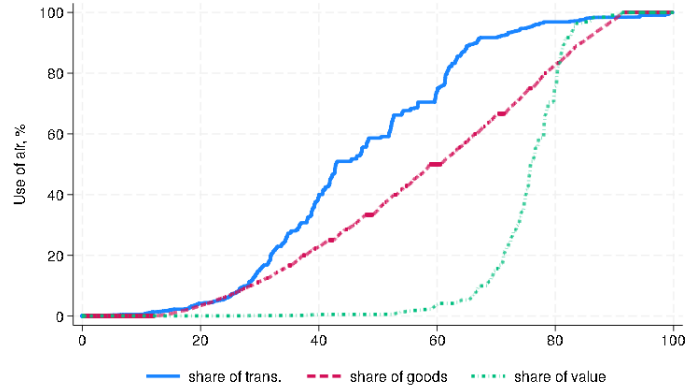
Note: Histograms display the distribution of air intensity across goods exported in Colombia using HS6-digit classification. In Panel A, we control for good’s unit value. In Panel B we control for city fixed effects, in addition to unit value. Air intensity is computed as the share of transactions within good that use air. Data Source: DIAN and author’s elaboration.

different shares, but it consistently lies between zero and one for all measures. This persistent mix suggests that firms do not uniformly select a single transport mode along any dimension, even after controlling for detailed observable characteristics.

B.3.3 Air Intensity and Financial Development: Cross-Country Evidence

To underscore the role of air shipment in explaining the link between financial development and specialization across countries, we show that countries with low financial access tend to specialize in goods with high air share. We first use the Colombian transaction data to compute the average air share by HS6 good, and then compute a country-level *air intensity* by averaging the HS6-level air intensity weighted by country-level HS6 exports. Figures B.6a and B.6b show average air intensity by decile of financial access and financial development, respectively, across country-years. In both cases, air intensity tends to be higher among countries with lower financial development. The relationship is non-monotone: air intensity peaks around the third decile before declining through the middle of the distribution, then rises again at the top decile. The uptick at the top is consistent with the fact that the most financially developed economies tend to specialize in high-value goods that are shipped by air regardless of financing conditions. Colombia sits at the sixth decile of both distributions,

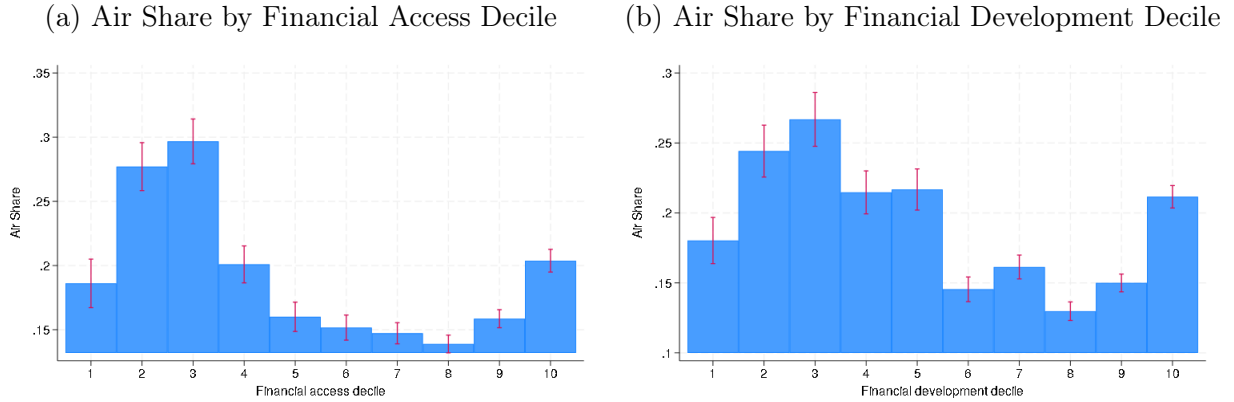
Figure B.5: Air proportion of air shipments using different measures



Note: The vertical axis shows the proportion of air usage. The blue solid line shows the share of transactions, the dashed red line shows the share of goods, the dashed-dotted green line shows the share of value. Data Source: DIAN and author's

in the range where air share is declining with financial development.

Figure B.6: U-shaped Air Share and Financial Development Across Countries



Note: Each bar shows the average air share by decile of financial development (left panel) and financial access (right panel), with 95% confidence intervals. Air share is computed as the share of export value shipped by air, using data from UN Comtrade. Financial development and financial access indices are from the World Development Indicators ([World Bank, 2024](#)). Sample covers country-year observations from 2008 to 2019.

Appendix C Model

C.1 Model calibration

Time-to-build The model implies that the total variable wage bill (excluding the fixed cost) in region o , sector j is given by:

$$w_o L_{oj}^v = \sum_d \sum_m \frac{\sigma - 1}{\sigma} (R_{oj})^{-(\bar{\tau}_j + \tau_{odjm})} \pi_{odj} X_{odj}$$

and the total debt required to finance variable labor is given by:

$$D_{oj} = \frac{1}{R_{oj} - 1} \sum_d \sum_m \frac{\sigma - 1}{\sigma} X_{odj} s_{odjm} [1 - (R_{oj})^{-\bar{\tau}_j - \tau_{odjm}}].$$

So that we can calibrate τ_j^0 to match the average sectoral debt-to-sales ratio in the data:

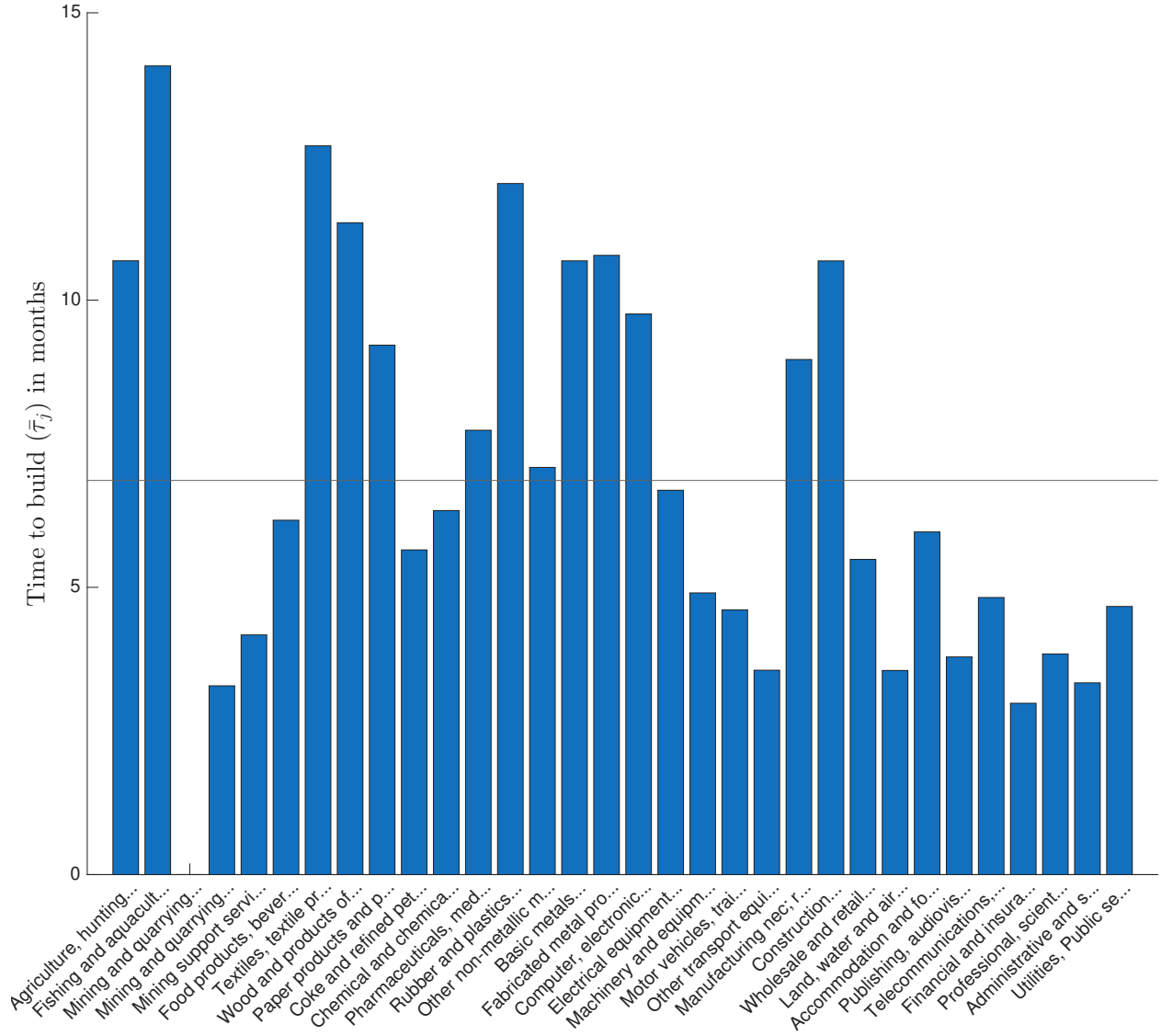
$$\frac{D_{oj}}{X_{oj}} = \frac{\frac{1}{R_{oj}-1} \sum_d \sum_m \frac{\sigma-1}{\sigma} X_{odj} s_{odjm} [1 - (R_{oj})^{-\bar{\tau}_j - \tau_{odjm}}]}{\sum_d \sum_m X_{odj} s_{odjm}} \quad (\text{C.1})$$

Given data on $\frac{D_{oj}}{X_{oj}}$, trade flows, mode shares, interest rates and τ_{odjm} , we use equation (C.1) to calibrate $\bar{\tau}_j$.

Figure C.1 plots our sectoral estimate of $\bar{\tau}_j$ by sector. Figures C.2 and C.3 show how it also correlates to the Rajan and Zingales measure of external finance dependence⁴¹ and air intensity respectively. Finally, Table C.1 correlates our time-to-build measure with the one constructed in Antràs and Tubdenov (2025), as well as with the share of transactions shipped by air. We find a positive correlation across measures and a similar relationship with air shipping intensity, which supports the fact that sectors with longer production lags face higher working capital needs, increasing the predominance of air shipping.

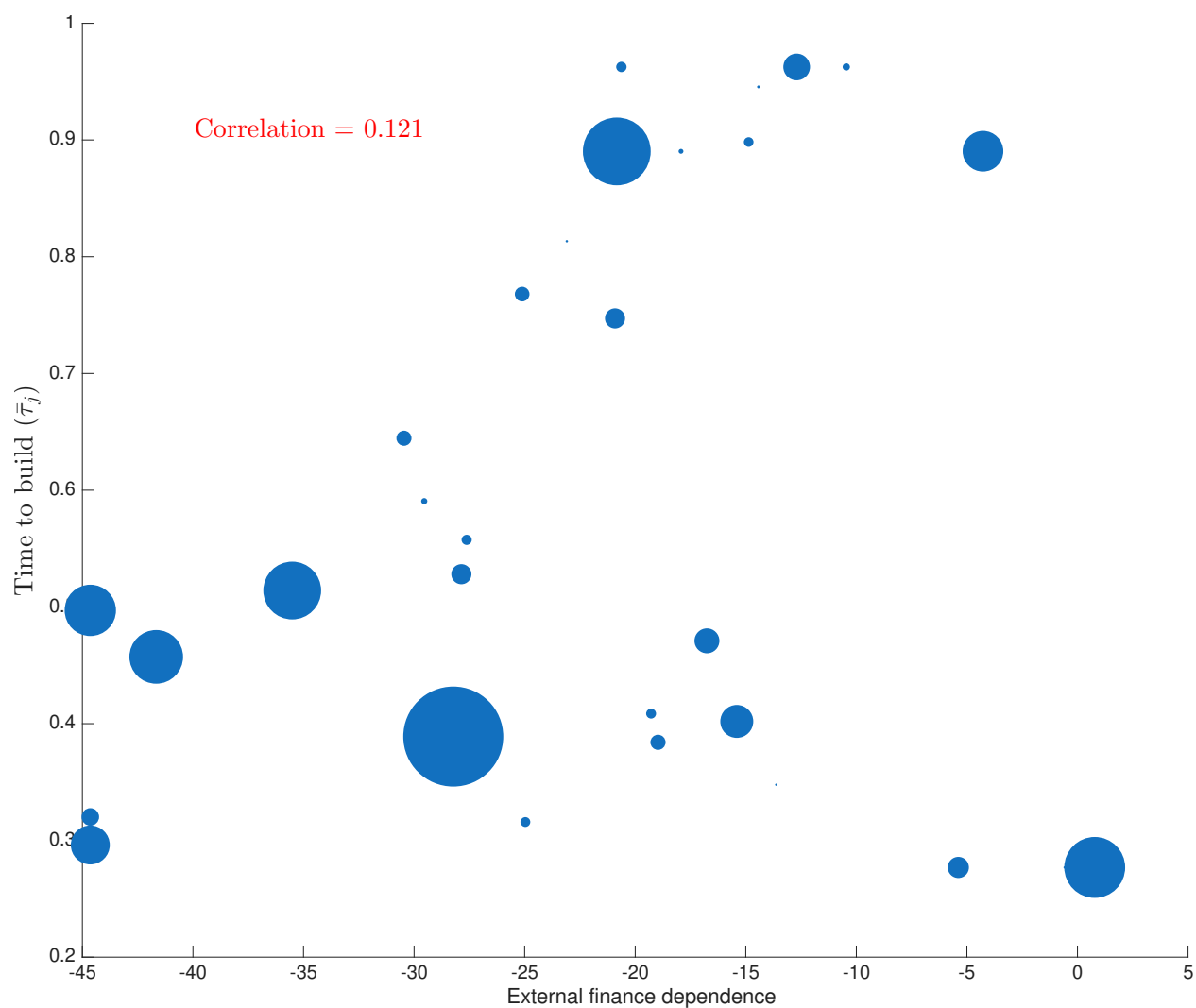
⁴¹The external finance dependence measure follows the methodology of Rajan and Zingales (1996). We construct capital expenditures at the firm level as the change in the real capital stock net of depreciation, using a perpetual inventory method. Cash flow is proxied by total sales. External finance dependence is computed as cumulative capital expenditures minus cumulative cash flow, divided by cumulative capital expenditures over the sample period. The sector-level measure is obtained as the median of this ratio across firms within each ISIC industry.

Figure C.1: Calibrated time to-build ($\bar{\tau}_j$)



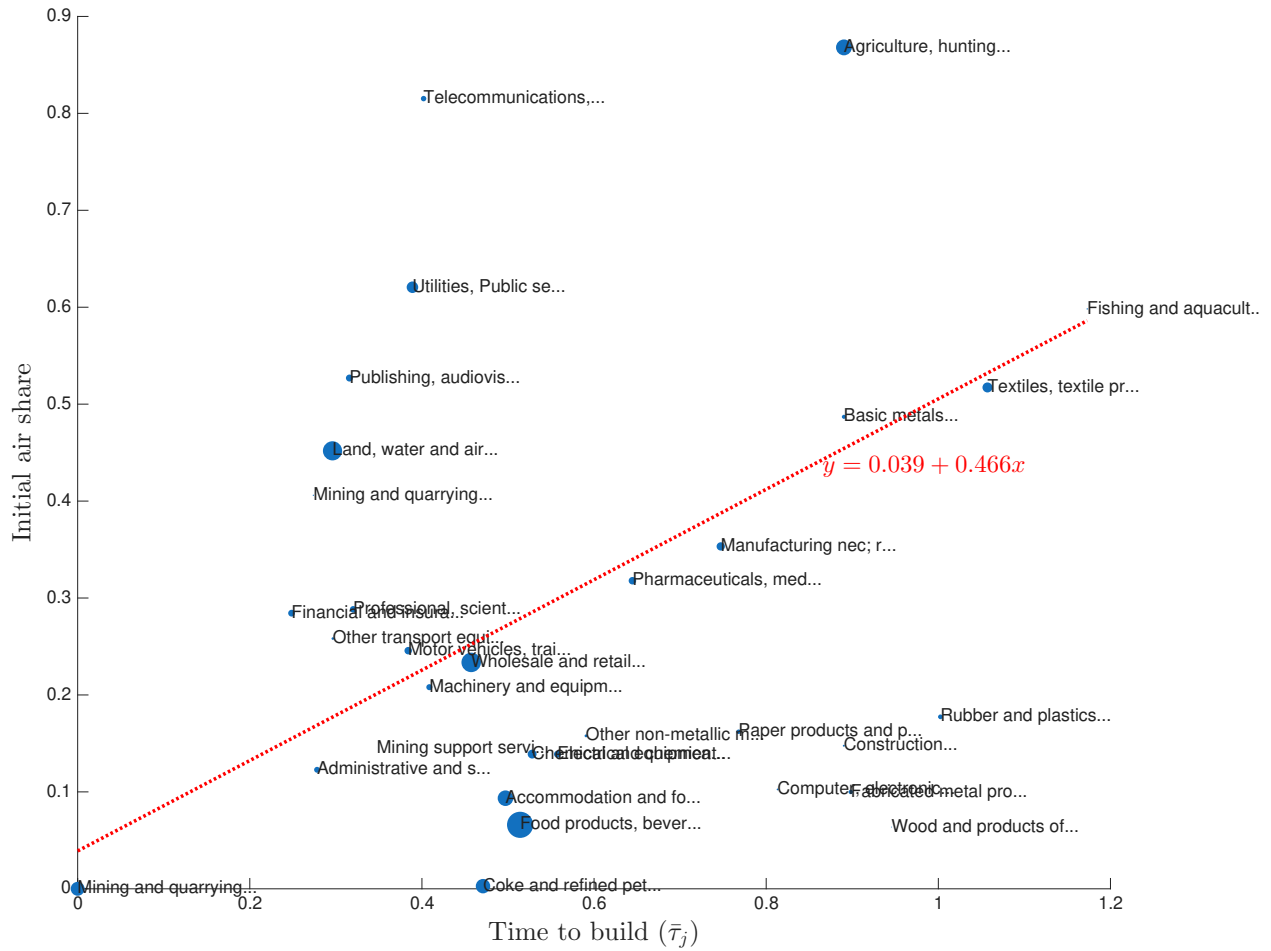
Notes: the figure plots the calibrated value of $\bar{\tau}_j$ obtained by matching the ratio of sectoral debt to sectoral sales in the data and the model.

Figure C.2: Other correlates of time to-build ($\bar{\tau}_j$)



Notes: the figure plots the calibrated value of $\bar{\tau}_j$ against the measure of external finance dependence.

Figure C.3: Time to-build ($\bar{\tau}_j$) and air intensity



Notes: the figure plots the calibrated value of $\bar{\tau}_j$ against sectoral air share in the data.

Table C.1: Time-to-Build Measures and Air Shipping Intensity

	(1) Our TTB	(2) Air share	(3) Air share
Antràs & Tubdenov's TTB	3.140*** (0.777)		1.211* (0.614)
Our TTB		0.518*** (0.133)	
Observations	23	23	23
R^2	0.522	0.509	0.147

Note: This table compares alternative time-to-build (TTB) measures and their relationship with air shipping intensity at the sector level. Column (1) regresses our TTB measure on the corresponding measure from Antràs and Tubdenov. Columns (2) and (3) relate sector-level air shipment shares to our TTB and Antràs and Tubdenov's TTB, respectively. All regressions are weighted by export value. Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.